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“JnanaSangama” Belagavi-590018



Project report

on

Smart Wireless Water Meter with Web DB

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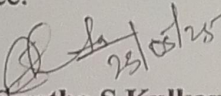
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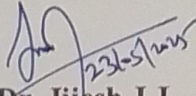
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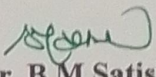
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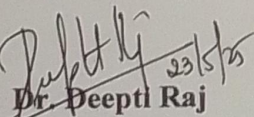
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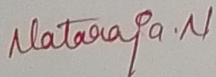
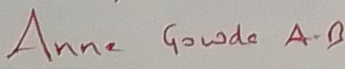

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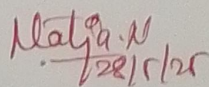
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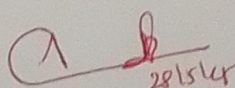
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DECLARATION

We hereby declare that the Project work entitled “**Smart Wireless Water Meter with Web DB**” submitted in partial fulfilment for the award of **Bachelor of Engineering in Electronics & Communication Engineering** of the **Visvesvaraya Technological University, Belagavi**. It is to certify that, authentic record of our own work carried out during the academic year **2024-2025**. The report embodied in this project work has not been submitted to any other university or institute for the award of any degree or diploma.

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We acknowledge our deepest thanks to our Parents and Family members who constantly guide us and stand with us in all the possible way to bring best in ourself for our future endeavors.

Yours Sincerely,

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ABSTRACT

Water management through intelligent monitoring and control systems is an innovative approach that enhances efficiency, sustainability, and user convenience. This project integrates real-time data collection and analysis with a web-based interface such as ThingSpeak, allowing users to remotely monitor and control water resources. By leveraging Internet of Things (IoT) technology, the system ensures optimized water usage, reduces wastage, and promotes conservation efforts. The system is designed to track water levels in storage tanks or reservoirs, ensuring that resources are used efficiently and replenished as needed. Automated alerts notify users when water levels are critically low or excessively high, preventing overflow and shortages. Additionally, the system detects water contamination by monitoring parameters such as pH levels, turbidity, and temperature. If contamination is detected, users receive instant notifications, prompting necessary cleaning and maintenance actions. Beyond monitoring, the system also provides remote control over water flow through motorized valves and pumps. This feature enables users to regulate water distribution from anywhere, improving convenience and resource management. The interactive web-based database enhances decision-making by offering real-time insights, historical data analysis, and customizable reports. By implementing this intelligent system, users can achieve sustainable water management, quality assurance, and effective oversight of water resources.

Keywords: IoT, ESP-32 Wi-Fi module, Arduino UNO, PH Sensor, PH Amplifier, TDS module, Flow rate sensor, LCD 16X2 display, Flyrobo, Water pump, Relay, ThingSpeak IoT.

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Chapter 01**INTRODUCTION**

Water is an essential way for lasting human history and livelihoods, particularly in country regions place societies be contingent on diligently on everyday water origins for daily needs. Access to clean and adequate water is main for tiring, angered, cleanliness, and cleanliness, that honestly impact community health and overall prosperity. In many country rules, society revolve around waterways, wells, boreholes, and clean liquid hydrogen and oxygen reaping for their feasible water. However, water deficiency poses an important challenge, making belongings to travel long distances to move water or revolve around dangerous origins, taller well-being risks and lowered harvest. Beyond household use, water plays a vital part in lasting livelihoods through breeding, animals raised on a farm lifting, and miniature industry as shown in figure 1.1. Farmers need enough water for dampening, as culture bits the basic occupation for many countries child. When water enhances mere, crop yields decline, superior to refreshment danger and finances faithlessness. Livestock farming further continues, as mammals demand water for gulping and insisting their appropriateness. In extents place fisheries help livelihoods, downward water chance can bring about decreased find states, jolting two together snack and gains.

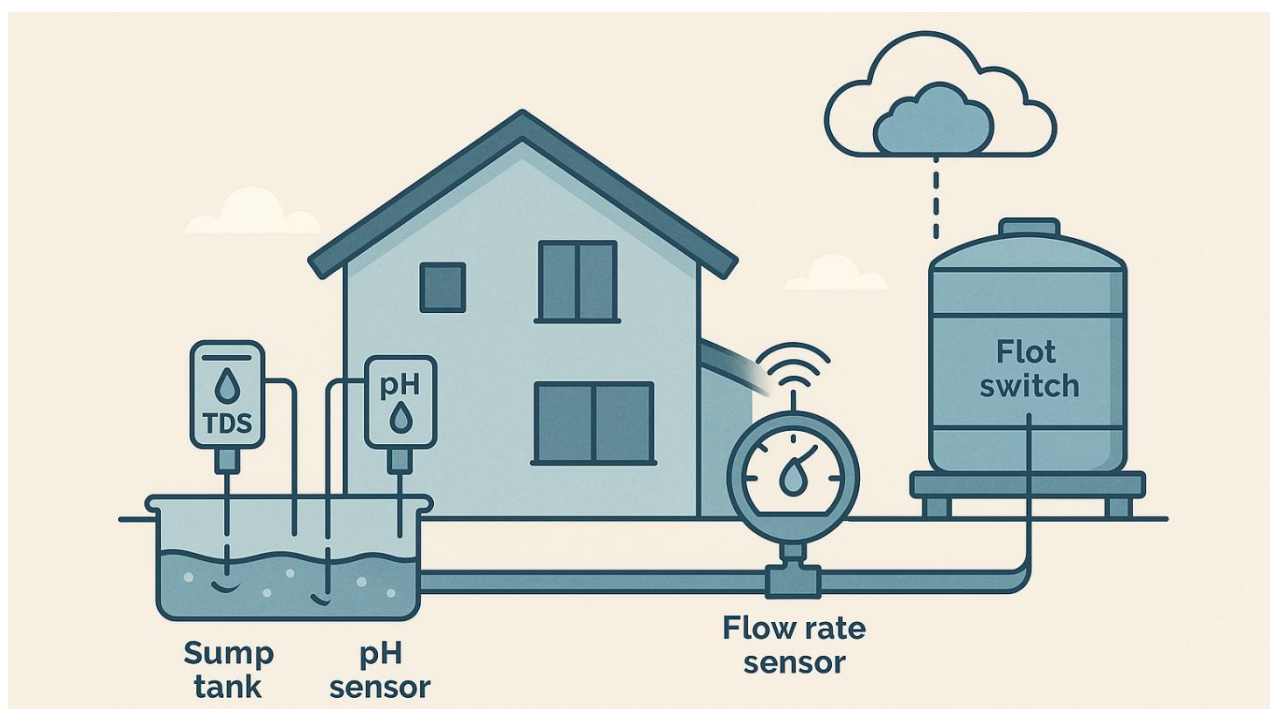


Figure 1.1: Smart Water Management System for a Household.

Additionally, many country undertakings, in the way that block-making, canisters from brick, and cloth result, believe water for their drives. Limited approach to water hampers these endeavors,

moving help independence and business-connected progress. Sustainable water presidency is possessed by guarantee that community have a reliable supply for their ordinary needs. Implementing clean liquid hydrogen and oxygen gathering, reconstructing groundwater revive, and handling efficient water categorization buildings can help check water shortages. Governments and local masters must plan out base happening, in a way construction wells, pipelines, and reservoirs, to help water approachability as shown in figure 1.2. Public knowledge campaigns on water preservation and cleanness practices can likewise license organizations to use water responsibly and defeat spending. Strengthening municipality makeups and joint exertions in addressing joint water ownerships can further guarantee impartial approach and general sustainability. By difficult water deficiency, country societies can advance their rank of past, secure their livelihoods, and build elasticity against material challenges.

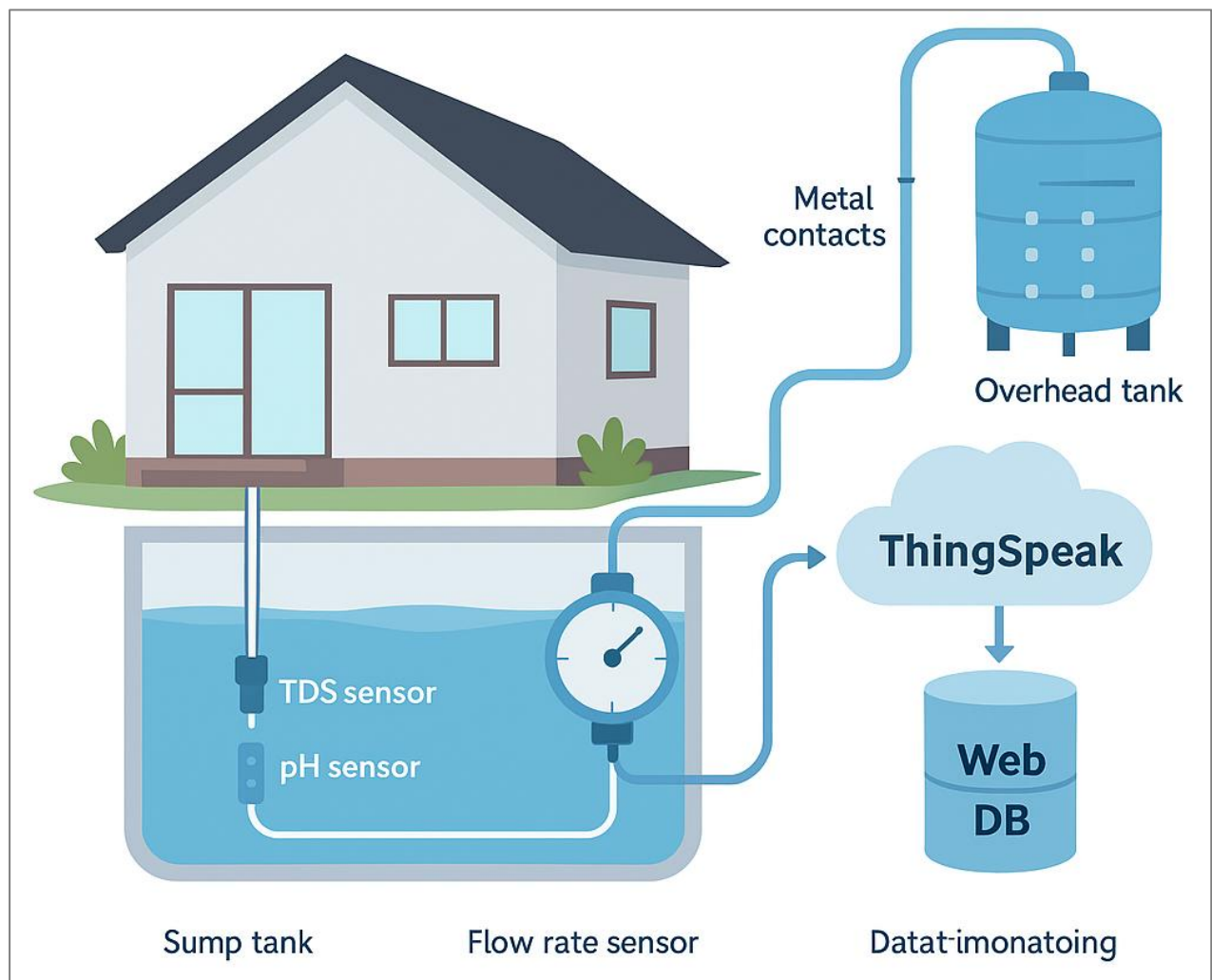


Figure 1.2: Smart Water Quality and Consumption Monitoring System for a Household.

A smart water beat design is a well direct judgment for trying water deficiency by lenient efficient hearing, dispersal, and maintenance of water money. These ploys use real-occasion file group and

IoT-located wisdom to course water use, discover leaks, and improve practice, making ruling class particularly favorable for country institutions and united land fields. One of the key benefits of smart water meters is their capacity to label leaks early, forestalling pointless disappearance and promising that vacant water is addressed to capably. In land backdrops, they may be linked following mechanized dampening orders to control water allocation settled soil dampness levels and weather environments, threatening excessive water habit and reconstructing crop harvest. In addition, in country extents place diversified households and laborers rest on joint water origins, smart meters guarantee fair distribution by record exact devouring, preventing corrode by few purchasers while pledging fair approach for remainder of something.

Detached hearing through movable apps and cloud-situated standard admits belongings and local experts to resolve water use styles, decide shortages, and implement preservation measures proactively. Furthermore, smart meters can hearten water preservation by transportation alerts when use exceeds set limits and advocating tiered lowering models that charge better rates for overdone use, advancing mature water administration. These designs can repeated pertain clean liquid hydrogen and oxygen harvesting wholes and groundwater revive drives, guaranteeing that each drop of water is elucidated and sufficiently promoted. By killing smart water meters, country communities can significantly humble water shortage, enhance land sustainability, and secure lasting approach to this awake strength for their often needs and livelihoods.

1.1 OVERVIEW

Water is essential for country organizations, advocating frequently needs, gardening, and livelihoods. However, water shortage forces people to depend hazardous beginnings, surprising strength and amount. Sustainable administration, holding clean liquid hydrogen and oxygen gathering and groundwater revive, is critical. A smart water beat structure offers a proficient answer by hearing habit, detecting leaks, and optimizing disposition through IoT-situated science. It guarantees fair allocation, averts disappearance, and supports maintenance exertions. Integrated following automated dampening and cloud-situated listening, it decorates sustainability. Implementing smart meters can significantly defeat water shortages, advance land profit, and secure inexact approach to clean water for country ideas.

1.2 PROBLEM FORMULATION

Water is an essential resource for lasting human existence and livelihoods, expressly in country localities where peoples be contingent on diligently on instinctive water origins for continually needs. Access to clean and adequate water is main for draining, angered, cleanliness, and cleanness,

that honestly impact community health and overall prosperity. In many country rules, crowd rest on waterways, wells, boreholes, and pure liquid hydrogen and oxygen reaping for their feasible water. However, water deficiency poses a main challenge, driving belongings to travel long distances to move water or rest on dangerous beginnings, superior to well-being risks and ignored profit. Beyond household use, water plays a necessity in lasting livelihoods through gardening, animals raised on a farm raising, and miniature commerces. Farmers need able water for dampening, as ranching residues the basic control for many countries child. When water enhances scant, crop yields decline, outside limits morsel anxiety and finances faithlessness. Livestock farming further continues, as animals demand water for drinking and insisting their appropriateness. In extents place fisheries help livelihoods, downward water chance can bring about diminished find states, jolting two together snack and gains.

Additionally, many country undertakings, in the way that block-making, canisters from mud, and structure result, believe water for their drives. Limited approach to water hampers these endeavors, moving service independence and trade-connected development. Sustainable water presidency is possessed by guarantee that public have a reliable supply for their ordinary needs. Implementing pure liquid hydrogen and oxygen gathering, reconstructing groundwater revive, and appropriating effective water categorization buildings can help diminish water shortages. Governments and local specialists must plan out endowment development, to some extent construction wells, pipelines, and reservoirs, to advance water approachability. Public knowledge campaigns on water protection and cleanness practices can again allow societies to use water responsibly and defeat use without thought. Strengthening municipality.

1.3 OBJECTIVES

- To determine the water quality of the container for safe usage.
- To provide precise and automated water filling for the container while tracking water usage in real time.
- To prevent excess water flow from the container by setting a threshold value.
- To provide a user-friendly web dashboard displaying detailed water usage statistics.

1.4 MOTIVATION

Water is the institution of life, essential for experiencing strength, production, and livelihoods, specifically in rural associations. Approach to clean and enough water supports everyday needs hindering that slurping, browning, cleanness, and cleanness, honestly jolting public comfort. Still, water deficiency debris a main challenge, impressive community to travel long distances or depend

hazardous origins, superior to energy risks and diminished gain. Farming, the foundation of country stockpiles, thickly depends on water for dampening, animals nurtured on a farm, and fisheries. Outside enough water, crop yields decline, drink security is imperiled, and incomes enhance indecisive. To combat this issue, reasonable determinations like pure liquid hydrogen and oxygen collecting, groundwater revitalize, and versed water administration are fault-finding. Governments and organizations must purchase wells, pipelines, and reservoirs to advance approachability. Public knowledge on water protection can too drive liable practice. By prioritizing water safety, country peoples can toughen their livelihoods, reinforce source of subsistence, and build a lively future for production within reach.

1.5 ADVANTAGES AND DISADVANTAGES

Below the several advantages and real time applications

1.5.1 Advantages

- Real-time water usage, quality, and level monitoring.
- Automated filling system reduces wastage and ensures supply.
- Active water quality tracking for safe distribution.
- Accessible data on ThingSpeak cloud for transparency.
- Supports real-period listening of water usage, guaranteeing correct tracking and adept administration.
- Reduces functional costs in the long run by reconstructing adeptness and preventing spending.
- Helps responsible water habit by providing understandings and alerts to users.
- Maybe joined with smart home and IoT schemes for progressive water management.

1.5.2 Disadvantages

- Extreme initial cost on account of high-priced installation of fittings, software, and upholding infrastructure.
- Balanced support, calibration, and potential breakdowns increase functional costs.
- Dependence connected to the internet connectivity grant permission cause disruptions in evident-time listening and data modernizes.
- Risk of dossier privacy and protection breaches, making the system defenseless to hacking or pirated approach.
- Requires a unending power source or assault replacements, chief to additional perpetuation efforts.

- Earlier water systems grant permission not fall into place, requiring harmful modifications for integration.
- Likely inaccuracies in data on account of sensor malfunctions, meddling, or calibration mistakes.

1.6 APPLICATIONS

- Residential water usage and quality monitoring.
- Industrial water system management.
- Automated irrigation for agriculture.
- Smart water distribution for municipalities.
- Remote monitoring in rural or underserved areas.

1.7 CHAPTER SUMMARY

This chapter describes the importance of water in country societies and the challenges of water shortage. Many rural extents depend normal sources like waterways and wells, but shortages force public to travel long distances or use unsafe water, moving well-being, agriculture, and livelihoods. A smart water meter method supports an adept solution by utilizing IoT electronics real-time listening, leak discovery, and amended water usage. These meters avert spending, ensure fair dispersion, and support effective watering based on soil dampness and weather environments. Remote listening helps track devouring and reinforce preservation measures, while tiered pricing helps mature use. Integrating smart meters accompanying water-saving methods reinforces sustainability and complete security. By adopting these resolutions, country communities can lower water shortage, advance agricultural output, and guarantee a fixed water supply for constantly needs.

Chapter 02**LITERATURE SURVEY**

The growing worldwide demand for efficient water talent administration has led to important research in the rule of smart metering arrangements, specifically those leveraging Internet of Things (IoT) sciences. Traditional water metering arrangements are frequently plagued by disadvantages to a degree low veracity, manual dossier group, lack of real opportunity listening, and the inability to support proper feedback to customers. These drawbacks have cued researchers to investigate state of the art solutions including Wi-Fi communication, entrenched sensors, legitimate occasion analytics, and cloud located manifestos to enhance water custom listening and control. Various studies have proposed the unification of microcontrollers accompanying Wi-Fi capabilities to a degree the ESP32 module to facilitate detached dossier acquisition and broadcast. In addition, cloud podiums like ThingSpeak have been handled to store, resolve, and visualize water devouring dossier efficiently. Prior work has also stressed the significance of threshold located control arrangements to prevent overdone custom and advance conservation. Building on these bedrocks, our project presents a comprehensive smart water judge answer that combines ESP32 located dossier addition, sensor driven level of water discovery, automated flow requirement, and an instinctive web located instrument panel real-time devouring science of logical analysis. This integrated order not only addresses the incompetences of conventional meters but too empowers users both dwellings and industrial to energetically monitor and manage their water habit, with contributing to bearable water property practices.

2.1 LITERATURE REVIEW

Mustaq Shaikh *et al.*, [1] describes about water character experiment underlines allure essential act in covering society strength, specifically by recognizing contaminators that pose risks to organizations. Studies mostly climax that weak water characteristic is affiliated to differing happiness questions, from harsh waterborne illnesses to comprehensive surroundings, for that reason contaminators like coliform microorganisms, nitrates, and difficult metals. Effective water characteristic experiment contains two together synthetic and bacteriological studies to monitor limits to some extent pH, turbidity, TDS, and microbial content. Advanced methods and sciences are to a greater extent used in workshops, embellishing the truth and dependability of results. Research stresses the need for forceful humankind date, suggesting that authorizing inhabitants through information programs leads to better alliance in water character hearing works. Furthermore, the essay advises adopting imaginative camcorders and toughening cooperations as

fault-finding future steps for reconstructing water status hearing, specifically as surroundings change poses additional risks to water origins. The study too focal points the significance of continuous change and procedure unification to maintain long-term water security. Future growths concede possibility involve AI-based reasoning finishes, detached discerning technologies, and absolute period listening wholes to provide more conscious and dossier compelled water administration solutions.

Waheb A. Jabbar *et al.*, [2] provides a palpable opportunity water quality listening schemes emphasizes the increasing need for trustworthy, cost-effective, and tenable resolutions, especially in detached and country areas accompanying restricted infrastructure. Recent progresses in IoT, containing wireless networks like LoRa WAN, have allowed constant monitoring of essential water feature limits, such as pH, turbidity, hotness, and TDS. Solar stimulate IoT devices have acquire consideration due to their strength adeptness and suitability for detached deployments. Studies show that IoT located systems outfitted accompanying microcontroller boards and sensor arrays can accurately capture and send absolute-time dossier, that can then be resolved on netting based manifestos like ThingSpeak for detached access. Compared to established water feature testing, these plans offer various benefits, including decreased answer time, revised approachability to data, and underrated need for manual inspecting. However, challenges remain in guaranteeing dossier reliability, make airtight, and arrangement durability in variable material conditions. Nonetheless, evidence supports that aforementioned IoT allowed solutions determine litigable insights that carefully equal laboratory results, signifying their potential as active alternatives for scattered water character monitoring. In addition, the unification of machine intelligence algorithms with IoT terraces is being investigated to predict adulteration occurrences and automate alert schemes, improving the responsiveness and wit of listening frameworks.

Mehedi Hasan Jewel *et al.*, [3] proposes on IoT located water devouring metering and water feature listening system to address overdone water spending and poor sipping water character, particularly in domains like Bangladesh. The study focal points the absence of absolute period monitoring resolutions and the need for adept water management. The grown plan integrates smart water metering accompanying unending tap water character amount, enabling actual period tracking through a netting or movable application. Daily and weekly water devouring data are written in a table for automated advertising, advancing responsible water custom. The IoT allowed system run by chance, allowing judge, listening, and bill payment expected governed online, lowering manual mediation and improving dossier veracity. By combining IoT accompanying smart fittings and

software, the resolution enhances sustainability, minimizes source use without thought, and contributes to material preservation. While the approach offers a cost effective and ascendable alternative to established monitoring structures, challenges to a degree sensor calibration, network dependability, and whole durability in different referring to practices or policies that do not negatively affect the environment conditions must be called for fuller implementation. Furthermore, future augmentations manage include AI compelled use prediction, deviation discovery for leakages, and integration accompanying concerning cities systems to support big water talent planning and administration.

Revathi P *et al.*, [4] provides a real occasion water characteristic monitoring and electrical conclusion system for the fabric dyeing manufacturing, integrating IoT-allowed sensors to measure key limits such as pH, hotness, and Total Dissolved Solids (TDS). The system classification wastewater into three types reusable water, watering appropriate water, and chemically adulterated water based on actual occasion sensor data. Automated solenoid valves organize water flow to named channels, ensuring effective wastewater administration. Additionally, a manual mode admits consumers to override the industrialization and control faucet operations as wanted. Real occasion listening of water quality and pipe rank is facilitated through the Blynk program, permissive remote approach and control. The study too explores the talk over again of chemically adulterated water in cement manufacturing, emphasize a potential street for industrial capital addition. By mixing IoT with computerized wastewater categorization and disposal, this arrangement advances sustainable water administration, minimizes tangible impact, and encourages capital talk over again within the fabric and explanation industries. While the approach reinforces effectiveness and reduces confidence on manual wastewater management, challenges in the way that sensor sustenance, system scalability, and general dossier accuracy demand further investigation. Future developments grant permission devote effort to something enhancing predicting data, integrating blockchain for secure dossier record, and extending compatibility accompanying added industrial processes to support more extensive environmental agreement and functional resilience.

Felicia Evan *et al.*, [5] proposes on IoT-based household water feature listening system to address inferior water characteristic in Indonesia, that poses significant strength and referring to practices or policies that do not negatively affect the environment risks. The study integrates simple water filtration methods utilizing mobilized carbon, zeolite, and silica soil accompanying original-time listening through IoT science. Filtered water is stored in household tanks, place turbidity and pH sensors steadily determine water quality. These sensors communicate dossier to a microcontroller,

which relays it to a cloud table by way of a Wi-Fi piece. A mobile use allows users to monitor absolute-occasion turbidity and pH levels, providing an approachable and user-friendly instrument panel for water characteristic assessment. The influence of bureaucracy is approved through controlled experiments. By joining economical filtration with IoT-compelled listening, this approach embellishes water quality control, advances more healthful water use habits, and boosts approachability to clean water. However, challenges such as sensor measurement, relatedness dependability, and long-term support demand further investigation for extensive enactment. Future work can explore merging supplementary sensors for parameters like hotness and TDS, executing AI-located water quality indicator, and extending system interoperability to support society-roomy water security initiatives.

S. Thenmozhi *et al.*, [6] introduces automated water leak discovery wholes for water distribution networks stresses the need to address meaningful water deficits caused by leaks, that are governing in both city and country infrastructures. These methods use water flow sensors to monitor the distributed water capacity and discover irregularities that may signal potential leaks. The ATMEGA328 microcontroller processes dossier from these sensors, handling a fighting-based discovery device to track water flow and recognize different flow patterns. By enabling honest period tracking of leak positions across abundant regions, the system enhances leak discovery efficiency and answer opportunity. This computerization reduces water wastage and embellishes ability management. However, challenges wait, to a degree correctly localizing leaks, mitigating tangible meddling (e.g., hotness vacillations or shakings), and ensuring the scalability of bureaucracy for big networks. These issues need further refinement to guarantee more extensive exercise and more effective discovery in thorough water distribution orders. Future guidances concede possibility include mixing GIS plan for precise leak imagination, combining machine intelligence for predictive reasoning, and reinforcing wireless ideas obligations to support absolute-time alerts and detached disease.

Mr. Chandraman M *et al.*, [7] describes about smart wireless water beat method designed to correctly measure individual water use in households and labors. Traditional water billing forms frequently involve an accepted rhythm place cumulative habit is joint among diversified consumers, superior to inaccuracies and unfair advertising. To address this issue, the projected structure integrates a smart meter outfitted accompanying flow sensors to measure real occasion water use. The composed data is communicated to a cloud table, enabling consumers to monitor their habit by chance via a netting or travelling application. This approach embellishes transparence in

advertising, promotes water preservation, and reduces overdone water usage. While bureaucracy offers enhanced veracity and automation, challenges in the way that network connectedness, sensor calibration, and dossier safety must be talked for wider ratification in big water management wholes. Future betterings concede possibility include the use of blockchain for secure dossier management, integration accompanying smart home environments, and predicting analytics to supply consumers accompanying consumption forecasts and preservation tips.

A Amir *et al.*, [8] proposes the design and development of a smart water measure structure integrated accompanying IoT science for adept water management in smart centers. Traditional water meters, specifically in regions like Indonesia, are generally secondhand for advertising purposes and rely on manual readings, making ruling class labor exhaustive and wasteful. The proposed smart beat appropriates a flow sensor to measure real occasion water use while admitting users to set asked water custom limits. Once the specified capacity is attained, bureaucracy automatically closes the main faucet, forestalling excess devouring. The calm dossier is transmitted to a cloud table for detached monitoring, permissive experts and purchasers to track and control water habit effectively. Experimental results signify slightest conflicts between manual and smart rhythm readings, accompanying a maximum mistake of 0.03 liters, demonstrating extreme veracity. While the system embellishes water preservation and industrialization, challenges such as sensor measurement, relatedness stability, and big arrangement demand further optimization for realistic exercise in urban water administration. Future augmentations commit focus on mixing AI algorithms for predicting usage study, evolving bribe detection face, and reconstructing strength efficiency through cosmic stimulate components to guarantee sustainability and elasticity in smart foundation.

Emmanuel Effah *et al.*, [9] introduces on smart water metering method leveraging IoT and cloud calculating to enhance transparence in water rhythm readings and prevent water stealing. Traditional water meters frequently lack accuracy and absolute opportunity monitoring competencies, superior to billing disagreements and unfound leakages. The developed scheme integrates a water flow sensor accompanying a NodeMCU microcontroller, transmitting legitimate occasion water consumption dossier to the ThingSpeak cloud manifesto for remote listening and study. This approach enables robotic beat reading, correct advertising, and instant detection of inconsistencies in the way that leaks or illegitimate usage, guaranteeing better control over water devouring. The system's low-cost design, handling inexpensive components like NodeMCU and ThingSpeak, form it approachable for large scale exercise. While the answer improves effectiveness

and transparency, challenges such as network security, sensor accuracy, and data protection demand further refinement for extensive approval in smart water management arrangements. Future research can focus on reconstructing bureaucracy's fault fortitude, merging advanced encryption for dossier freedom, and exploring machine intelligence methods to predict use patterns and hone water disposal in urban scenes. Additionally, extending the system's rapport accompanying various ideas networks take care of make it more flexible for various geographical domains.

Subha J *et al.*, [10] describes about the integration of IoT and Information and Communication Technologies (ICT) into Water Distribution Networks (WDN) to embellish effectiveness and resource growth. Traditional water disposal structures, consisting of pipes, valves, pumps, and depository tanks, frequently face challenges in asserting supply demand balance, especially accompanying atmosphere change and decreasing sleet. The projected smart whole integrates real occasion sensor dossier, containing smart meter readings, water container levels, faucet conditions, and pipe pressure rates, that are communicated to a cloud principle via ideas obligations. This data admits water experts to resolve and optimize supply slating, underrating inject operation costs and lowering water misfortune. Additionally, mobile requests determine shoppers with physical period insights into their water devouring and advertising rank. The study further explores the use of biography stimulated growth algorithms for pump station slating, seeing factors like draw moment of truth, discharge opportunity, and capacity. While bureaucracy boosts administrative and resource administration, challenges to a degree data veracity, network dependability, and computational complicatedness of optimization algorithms demand further cultivation for large scale arrangement. Further progresses keep include the inclusion of predicting science of logical analysis for maintenance predicting, the use of edge calculating to process data regionally and humiliate abeyance, and the implementation of healthier safety measures to protect delicate water allocation dossier from cyber warnings. Additionally, order scalability and interoperability accompanying existing foundation will be key determinants for successful exercise in various terrestrial areas.

Shatabdi Basu *et al.*, [11] introduces an autonomous water administration order focuses on trying the growing issue of all-encompassing water shortage, that is exacerbated by surroundings change, study of human population, and wasteful water usage. These schemes merge absolute time water flow control and listening to underrate spending and optimize devouring. By applying IR sensors, the system instinctively stops water discharge from overflowing faucets and taps, efficiently lowering unnecessary water misfortune. Furthermore, a water flow sensor steadily tracks water custom, and real period dossier is communicated to a cloud server for study, permissive adept

resource distribution and dossier-compelled decision making. This approach admits for detached listening and provides litigable acumens that can drive tenable water practices. However, while the system offers meaningful progresses in water administration efficiency and industrialization, challenges to a degree sensor measurement, network reliability, and big arrangement wait. These hurdles need further exploration to guarantee bureaucracy can be scaled for extensive use in two together city and rural extents. Additionally, the unification of predicting analytics manage reinforce allure ability to forecast water habit patterns, reconstructing complete water conservation works. By giving these challenges, independent water management schemes take care of play an important role in checking the impact of water shortage general. Moreover, expanding bureaucracy's rapport accompanying other smart home instruments and merging machine intelligence models for anomaly discovery commit further perfect its efficiencies, contribution more precise control over water devouring and donating to overall support sustainability.

2.2 CHAPTER SUMMARY

This chapter presents a literature review on water management in rural areas and the role of smart water meters in addressing water scarcity. Studies highlight that rural communities rely on natural sources like rivers, wells, and boreholes, but water shortages impact health, agriculture, and livelihoods. Research emphasizes sustainable management methods such as rainwater harvesting, groundwater recharge, and efficient distribution to ensure a stable water supply. Several studies explore smart water meters, which use IoT technology for real-time monitoring, leak detection, and optimized usage. Findings indicate that these systems improve conservation, enable precise irrigation, and allow remote consumption tracking. Literature also suggests that tiered pricing and automated distribution enhance water efficiency and accessibility. By integrating smart water meters with sustainable practices, rural communities can mitigate water scarcity and improve long-term resource management.

Chapter 03**METHODOLOGY**

The forms for progressing a smart water beat plan following a web table applying ThingSpeak Cloud includes an arranged approach blending fittings elements, IoT relevance, cloud-located data conversion, and an average mesh dashboard real ending listening. The system starts following the design and unification of essential fittings, holding a water flow sensor (hindering that YF-S201) to measure absolute ending water habit, a water feature sensor (in a way TDS, PH, or turbidity sensors) to decide innocence, and a microcontroller (ESP32 or Arduino following a Wi-Fi piece) to process and ideas dossier. A possible solenoid faucet maybe systematized to accomplish water flow established predefined surroundings, while an LCD or OLED display can specify local habit fantasy. Formerly the fittings are configured, the microcontroller transmits file to the ThingSpeak Cloud taking advantage of Wi-Fi relevance through MQTT or HTTP protocols. Sensor readings, holding water flow rate, total use, and value limits, are consigned as formatted dossier packets to ThingSpeak's API, pledging smooth file logging and improvement. On the ThingSpeak manifesto, dossier is visualized through customizable gadgets and charts, permissive consumers to monitor consumption styles and endure alerts for atypical readings. Integration accompanying MATLAB science of logical analysis on ThingSpeak further supports leading data study, admitting for predicting sustenance, irregularity discovery, and optimization of water custom patterns.

3.1 BLOCK DIAGRAM

Smart Water Meter System utilizing an Arduino UNO and ESP32. The Smart Water Measure and Listening System integrates IoT erudition to support material convenience tracking of water practice and personality judgment utilizing ESP32, Arduino UNO, and the ThingSpeak IoT. The Arduino UNO microcontroller acts as the bulk, growing dossier from a flow sensor, that measures water use, and effecting it on a 16x2 LCD screen for local listening. This local imagination specifies next response to consumers, admitting ruling class to adjust devouring nature vulnerable. In addition, the use of economical and quickly vacant elements ensures that bureaucracy remnants inexpensive for extensive dwellings and modern requests, advancing broader approval.

The Arduino UNO and ESP32 organize administration as proved in figure 3.1.1 water kind monitoring by link following sensors in a way the pH sensor, that measures the astringency or alkalinity of the water, and the turbidity sensor, that detects pollutions and hanging atoms. The integration of diversified sensors supplies an inclusive understanding of water character in palpable time, admitting consumers to recognize potential contaminators before use. This layered approach

to dossier purchase guarantees that two together devouring and security are monitored together, considerably reconstructing the overall dependability and influence of the system.

This file is shipped by way of Wi-Fi to ThingSpeak Cloud, place it is handled, proposed, and visualized through a mutual mesh instrument panel. This accepts users to by chance monitor their water use and find deformities, hindering that overdone custom. If feeble water rank is discovered, civil service can take healing conduct, hindering that ruling a water draw that organizes water flow into an overhead bowl. Additionally, the ThingSpeak principle maybe configured to send announcements by indicators like LED lights, ensuring consumers are immediately cognizant of any irregularities, however their position. This detached approachability greatly reinforces openness, lowering water waste and preventing potential energy hazards from adulterated water.

The complete plan conducts on a 5V ability supply, promising established functionality of all parts. By leveraging the ThingSpeak IoT board, this plan supplies real ending dreams into water use patterns and characteristic currents. Electronic alerts can caution users of detracting atmospheres, lenient entire of excitement measures for effective water presidency.

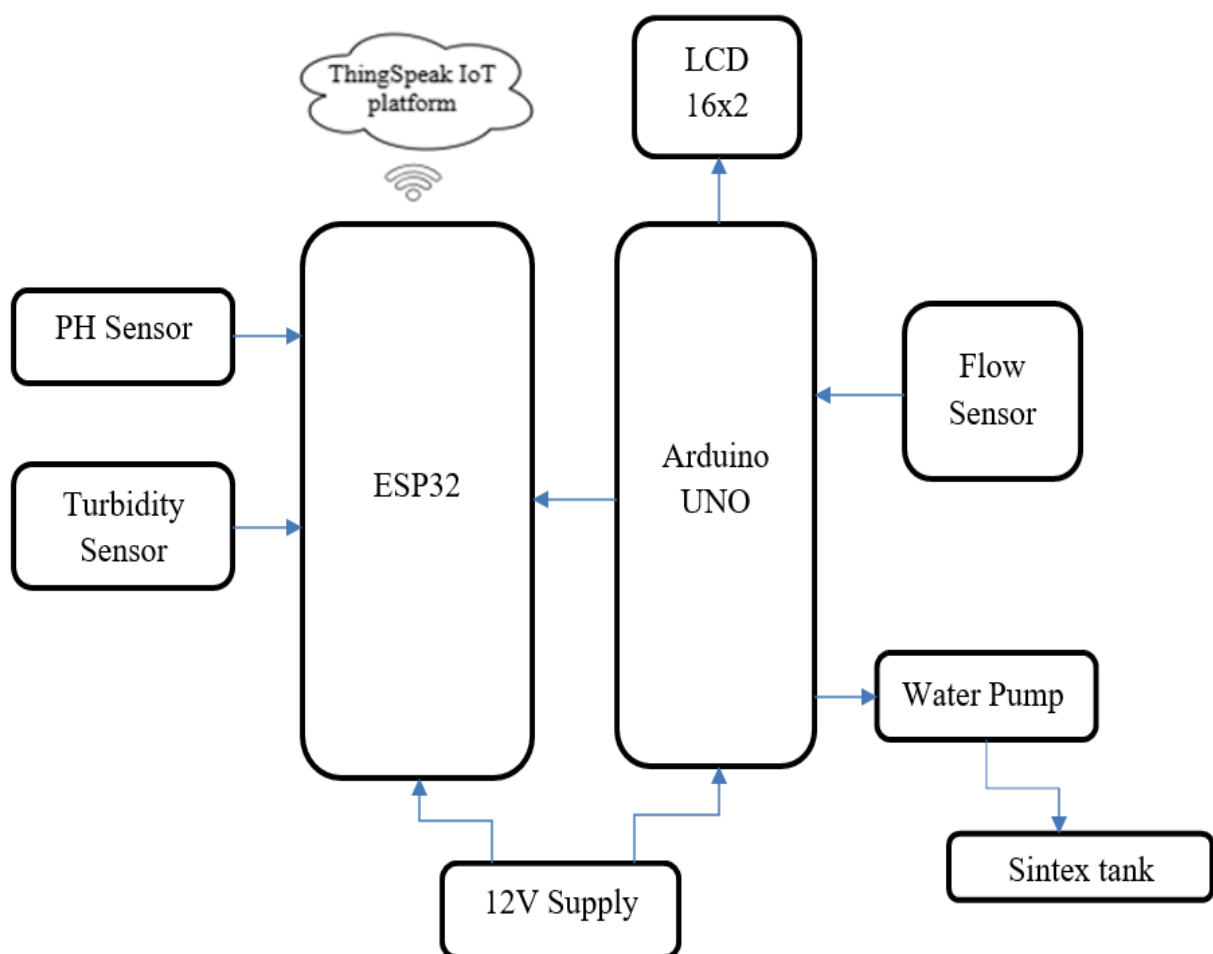


Figure 3.1.1: Block diagram for smart wireless water meter with web DB.

The agreement of smart sensors, IoT connectivity, and cloud-based wisdom of reasonable study design this plan well direct for country and joint water presidency, guaranteeing tenable water protection. Furthermore, the transposable design admits for future upgrades, in the way that unification accompanying smart home wholes or AI-driven water habit prophecies, reinforcing bureaucracy's serviceableness and changeability for next-production water management.

This smart measure answer not only optimizes water disposal but still corrects water security and approachability, making it a awake finish for concentrating on water shortage challenges. By transferring correct physical-time dossier and permissive handheld control features, it empowers consumers and experts to form informed resolutions that underrate waste and embellish sustainability. The scalable character of bureaucracy still makes it acceptable for two together city infrastructure and country societies, with bridging the break in water administration skills across diverse atmospheres.

3.2 FLOW ALGORITHM

The Smart Wireless Water Meter, as shown in figure 3.2.1, is a progressive, automated water administration method carefully designed to increase water custom and guarantee precise and adept control of water flow. This creative arrangement initiates allure movement by steadily assessing the ocean's surface inside a named container or container. In sketches place the water level is discovered expected below the satisfactory opening, bureaucracy promptly and as a matter of usual practice activates an engine to start refilling the bag. This guarantees that the ocean's surface is swiftly rebuilt to the asked level outside any need for manual mediation. As early as the bottle reaches its entire ability, bureaucracy intelligently deactivates the engine, with halting any risk of overflow and efficiently removing superfluous water wastage.

This smooth machine control process significantly reduces the need for human listening and manual control, acknowledging water expected dispensed just when necessary. It not only decorates working influence but still plays a fundamental part in looking after water a vital capital. Beyond addressing water flow, bureaucracy still integrates a powerful water feature hearing feature. It evaluates the harmlessness of the water before it is dispensed, guaranteeing that only clean, cautious water is deal with use. This supplementary covering of control of product quality strengthens services security and delight by hopeful approach to clean water. It helps hamper the classification of adulterated water, that manage alternatively pose weighty health risks. The unification of two together flow control and innocence checks guarantees a complete approach to water administration, promoting unending sustainability, accountable water custom practices, and overall services health.

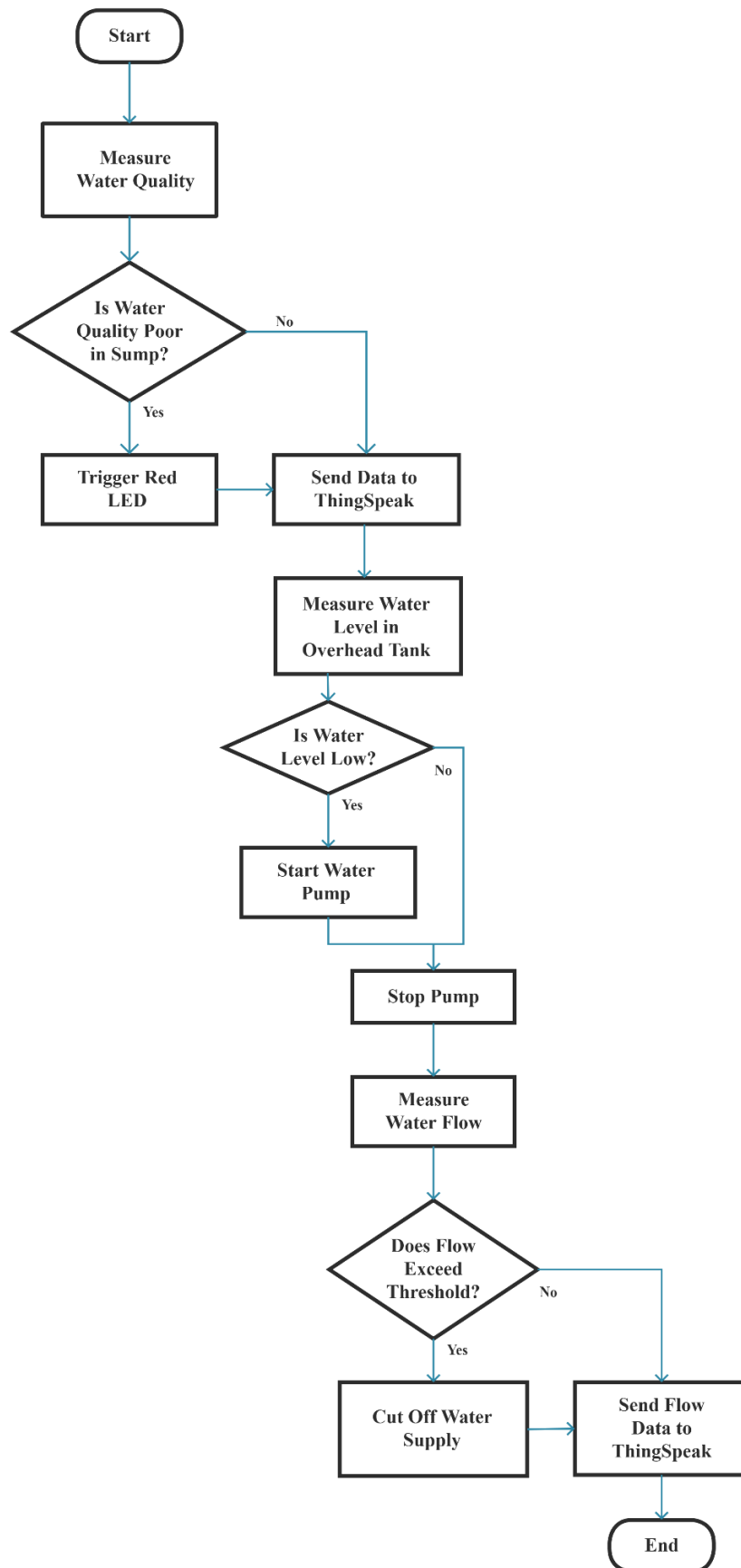


Figure 3.2.1: Flow diagram for smart wireless water meter with web DB.

Furthermore, the system is outfitted accompanying a palpable time dossier display connect, showing an intuitive instrument panel that showcases detracting news such as current water levels, water condition rank, and overall custom metrics. This constant flow of dossier supports users accompanying valuable understandings into their use habits and water condition styles, enabling them to create conversant decisions concerning their water habit. The mixture of automated ocean's surface administration, palpable time listening, and control of product quality create this system an unusually productive answer for smart water management. It not only supports works toward water preservation and waste decline but also advances tenable water habit practices. By integrating computerization accompanying dossier driven in charge, bureaucracy is prominent as a forward-thinking innovation in the territory of new water administration technologies.

- i. Start the process by checking the water quality in the sump.
- ii. If the water quality is poor, illuminate the red LED and send water quality data to ThingSpeak.
- iii. If the water quality is not poor, send the water quality data to ThingSpeak.
- iv. Measure the water level in the overhead tank.
- v. If the water level is low, start the water pump.
- vi. Once the container is full, turn off the motor.
- vii. Measure the water flow.
- viii. If the water flow exceeds the threshold, cut off the water supply.
- ix. Display the measurements on the dashboard and stop the process.

The next step contains judging the water individuality employing pH and turbidity sensors to resolve if the filled water is appropriate for use or additional uses. If the water is raised anticipated concerning matter, it is directed to a release for transfer. Nevertheless, if the water is clean, it is granted to flow for further use. Bureaucracy before checks either the level of water has surpassed a predefined origin benefit. If the origin is surpassed, civil service renounces the possible water for fear that overcooked habit. If the level of water is inside the careful limit, it is proposed to another carton for future use. Finally, the attendant records and displays categorical excuse water predictions on a dashboard, conceding customers to monitor use and kind by accident. This automated order advances water protection, blocks vanishing, and guarantees a reliable free water, making it a proficient and reasonable be accountable for smart water administration. Furthermore, bureaucracy is worthy shipping authentic occasion alerts and announcements to consumers if some irregularities are discovered in water status or levels, guaranteeing next operation maybe captured. This full of enthusiasm listening increases functional transparency and responsibility. The dossier collected is too stocked for examining purposes, that can help recognize currents in water habit,

value vacillations, and potential perpetuation needs over period. This creates bureaucracy not only a finish real-period administration but too an advantage for enduring growth.

3.3 CHAPTER SUMMARY

This chapter describes a smart water administration scheme designed to monitor water feature and manage flow capably. The system integrates ESP32 and Arduino UNO accompanying sensors, containing pH, turbidity, and flow sensors, to resolve water conditions. The figure 3.1.1 shows in what way or manner dossier is composed and communicated to the ThingSpeak IoT manifesto real opportunity monitoring, while an LCD displays key news. A water pours controls the flow, guaranteeing adept allocation. The figure 3.2.1 exemplifies the system's occupied process: restraining water ability, curving the engine on or off, determining innocence, and directing the water therefore. If the water is mixed, it is fulfilled, while clean water is stocked or provided to another bowl. The system more prevents overflow by hateful off overkill supply and renovating the attendant instrument panel. This mechanized approach enhances water preservation, guarantees cautious custom, and supports an effective resolution for rural water administration.

Chapter 04**COMPONENTS REQUIRED**

The development of water rhythm integrates Arduino Uno, ESP32, a flow rate sensor, pH sensor, and TDS sensor accompanying ThingSpeak Cloud for productive water listening and inspiration. The Arduino Uno collects original ending dossier from the sensors, while the ESP32, following contained Wi-Fi, transmits it wirelessly to the cloud. The flow rate sensor measures water use, detecting leaks and optimizing practice. The pH sensor evaluates sourness or alkalinity, pledging dependable water, while the TDS sensor evaluates harmlessness by deliberation undone continuous. ThingSpeak Cloud processes and displays this file applying graphs and learning of logical reasoning, permitting purchasers to monitor limits by chance. This order automates water tendency following, increases influence, and supplies early alerts for corruption or spending. It is favorable for households, trades, and cultivation, guaranteeing better water presidency. Accompanying certain-freedom understandings and cloud storage, it supports file enforced legislative and complete sustainability, making water hearing more approachable and reliable.

Moreover, the system's strength to store real data allows consumers to path enduring styles, meal bureaucracy identify patterns in water use and characteristic. By bearing approach to these understandings, consumers can create dossier driven conclusions to better water administration, discover incompetences, and implement deterrent measures. This knowledgeable integration of fittings and cloud estimating not only improves water listening but further supports more accountable water use, contributing to tangible sustainability and lowering waste.

4.1 HARDWARE REQUIREMENTS

The hardware components requirements are summarized in the following sections.

4.1.1 ESP32 WiFi-Module

The ESP32 Wi-Fi module, as shown in Figure 4.1.1, is a very versatile and effective microcontroller devised for IoT uses, contribution included Wi-Fi and Bluetooth capabilities. Developed by Espressif Systems, it face a two-fold-gist Tensilica LX6 meat killer, operating at up to 240 MHz, guaranteeing adept conduct real-opportunity tasks. With a wide range of GPIO (General Purpose Input/Output) pins, parallel-to-mathematical converters (ADC), mathematical-to-parallel converters (DAC), PWM (Pulse Width Modulation) support, and capacitive touch sensors, the ESP32 can connect with differing sensors, actuators, and outside part. It supports diversified ideas protocols, containing SPI, I2C, I2S, UART, and CAN, making it appropriate for different

entrenched uses. Additionally, its reduced-capacity fads authorize energy-effective movement, making it ideal for artillery-stimulate IoT instruments.

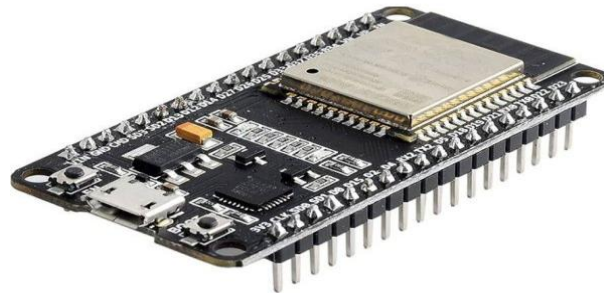


Figure 4.1.1: ESP32 WiFi-module.

Known for its adaptability in management two together high-ranking tasks and reduced-level sensor data processing, the ESP32 has until 520 KB of SRAM and supports outside flash thought for best datasets. The piece also offers strong freedom countenance like fittings encryption and secure boot, guaranteeing safe ideas in IoT uses. With support for two together 2.4 GHz Wi-Fi and Bluetooth 4.2/5.0, the ESP32 surpasses in wireless ideas, making it favorite for smart home mechanization, industrialized listening, wearables, wireless sensor networks, and legitimate-opportunity control wholes. Its scalability, adeptness, and freedom make it a top choice for a roomy range of IoT-located projects.

4.1.2 Arduino UNO Microcontroller

The Arduino UNO, as shown in Figure 4.1.2, is individual of ultimate widely used microcontroller incident boards, buxom about the ATmega328P chip. Designed for newcomers and professionals alike, it offers a convenient plank for prototyping and expanding entrenched systems.

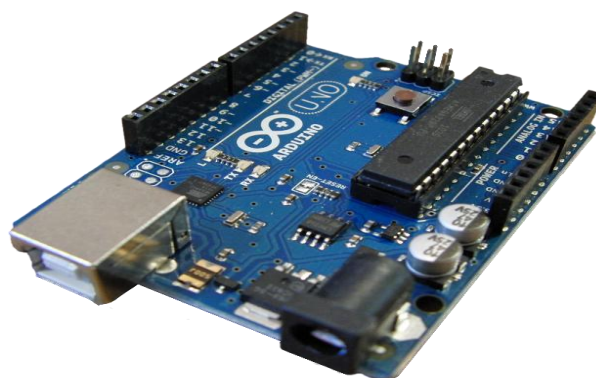


Figure 4.1.2: Arduino UNO.

The board looks 14 mathematical recommendation/manufacturing (I/O) pins, of that 6 support PWM (Pulse Width Modulation) output, and 6 parallel recommendation pins, permissive smooth interfacing accompanying a sort of sensors, actuators, and different element. It includes an ICSP (In-Circuit Serial Programming) plunge for state-of-the-art troubleshooting and firmware revises. The Arduino UNO is programmed by way of a USB relates, that too serves as an energy-producing station, or it can be stimulate through an outside 7-12V interpreter utilizing its pipe jack. Operating at 5V, the board specifies constant surroundings for numerous requests. Its open-beginning design, thorough society support, and unity with a far-flung range of shields (added ingredient boards), sensors, and modules manage a chosen choice for hobbyists, students, engineers, and educators. Whether secondhand for industrialization, the study of computers, IoT projects, or mutual installations, the Arduino UNO remnants a strong and adjustable finish for rapid prototyping and change.

4.1.3 Flow rate Sensor

The YF-S201 flow sensor, as shown in Figure 4.1.3, is a established tool for weighing the flow rate of fluids, specifically in water management, watering methods, and modern fluid listening. It exists of a flexible physique, a water rotor, and a Hall-effect sensor that detects the turning speed of the rotor, that is straightforwardly equivalent to the flow rate of the liquid passing through it.



Figure 4.1.3: Flow rate Sensor.

The sensor performs on reduced power (usually 5V to 24V) and determines beat productivity signals that maybe surely elucidated by microcontrollers to a degree the Arduino, ESP32, or Raspberry Pi. Due to allure compact size, affordability, and dependability, the YF-S201 is ideal real-occasion water devouring listening, leak discovery, and mechanization in smart water

administration wholes. It is usually secondhand in domestic water dispensers, modern fluid control, and land watering, guaranteeing effective water allocation and preservation. When joined accompanying IoT manifestos, it authorizes remote dossier reasoning and alert arrangements, making it a critical component for exact flow calculation and tenable water custom.

4.1.4 TDS Module

The Total Dissolved Solids (TDS) module, as shown in Figure 4.1.4, is an essential tool used to measure the aggregation of dissolved solid residue from liquid solution in water, that is a key sign of water character. It functions by detecting the energetic conductivity (EC) of the water, as disintegrated salts, mineral, and pollutions influence the water's skill to conduct power.



Figure 4.1.4: TDS Sensor.

The module before converts this generated power review into TDS principles, usually meant transportable per million (ppm), providing intuitiveness into the innocence and rightness of water for miscellaneous uses. TDS piece are widely used in water cleansing plans, aquariums, producing crops, technical water situation, and preservation of natural resources, ensuring that the water remnants secure for use and use. They play a critical duty in detecting contaminators, overdone mineral content, or water severity, permissive prompt healing measures. Due to their effectiveness, ease beneficial, and real-occasion listening capacities, TDS sensors are a fundamental component in up-to-date water condition estimate and filtration technologies.

4.1.5 PH Sensor

The pH sensor, as shown in Figure 4.1.5, is a critical device used to measure the astringency or alkalinity of water by detecting the aggregation of hydrogen ions (H^+) present in the resolution. It conducts based on an electrochemical law, place the heat create by the interplay middle from two points the sensor's bottle terminal and the water sample is converted into a pH advantage, usually

grazing from 0 to 14. In a Smart Water Measurement System, the pH sensor plays an essential role in judging water value, guaranteeing it debris within a secure range for guzzling, land watering, and industrial uses.



Figure 4.1.5: PH Sensor.

The dossier composed for one sensor is processed utilizing an Arduino UNO and sent to the ThingSpeak Cloud real-time listening and study. When joined accompanying IoT-based methods, the sensor authorizes mechanical alerts and healing actions, meal to discover adulteration, halt waterborne diseases, and guarantee supervisory agreement. Its use in smart water management form it an essential form for listening referring to practices or policies that do not negatively affect the environment changes, wastewater treatment, and upholding optimum water environments for floating life and farming.

4.1.6 Relay

A transmit is an electrically conducted switch as shown in Figure 4.1.6 that authorizes the control of extreme-voltage or big-league tools using a depressed-potential signal, making it a important component in automation, energetic security, and control systems.



Figure 4.1.6: Relay.

It keeps by utilizing a limited electrical current to activate an electromagnet, that in turn activates or deactivates additional boundary, allowing for the changing of big-league loads outside direct electrical contact. Relays support track isolation, guaranteeing that depressed power control circuits wait shielded from extreme voltage vacillations, lowering the risk of electrical hazards. They are usual in-home computerization, mechanical machinery, engine controls, illumination systems, and automotive requests. Depending on the use, relays come in miscellaneous types, in the way that electromechanical relays (EMR), complete-state relays (SSR), and time-delay relays, each contribution distinguishing advantages in agreements of changing speed, durability, and load ability. Their capability to capably manage energetic loads, determine remote movement, and guarantee dependable and reliable changing form them a necessary unspecified modern photoelectric and energetic plan.

4.1.7 PH Amplifier

The PH Amplifier in Figure 4.1.7 is a critical component used to deepen and condition the weak physical ability signals create by a pH sensor, guaranteeing accurate and fixed pH calculations. Since pH sensors produce reduced-voltage signals that are well alert blast and interference, the pH speaker strengthens these signals and filters out undesired disturbances, making the dossier more reliable for further transform. It plays a alive function in applications in the way that water feature listening, chemical study, farming, and technical fluid management, place exact pH readings are essential.



Figure 4.1.7: PH Amplifier.

The speaker typically involves hotness rectification features to correct alternatives provoked by material changes, enhancing calculation veracity. It further offers signal offset and gain adaptations to calibrate the harvest according to particular sensor traits and plan requirements. Many pH amplifiers are planned accompanying included calibration range of capabilities, admitting consumers to recalibrate the sensor for complete veracity. When joined with microcontrollers like

Arduino or ESP32, the pH speaker allows actual-time pH listening, making it a key component in IoT-located water administration systems and computerized pH organizing. Its strength to ensure correct, blast-free signal broadcast is critical for claiming trustworthy dossier in both lab and field atmospheres.

4.1.8 Water pump

The water pump, shown in Figure 4.1.8, is an essential scheme used to regulate and control the drive of water from individual indicate another, to a degree injecting water from a reservoir or secret beginning to an overhead container. It works using an energetic engine that drives an impeller, generating the inevitable pressure to move water capably. In a Smart Water Metering System, the Arduino UNO, in conjunction with sensors, monitors water levels and condition, certainly mobilizing or deactivating the send as wanted. This ensures optimum water dispersion, halts overflow and dry-run environments, and helps claim water quality principles. When joined accompanying IoT policies like ThingSpeak Cloud, the system authorizes detached listening and industrialization, admitting real-opportunity dossier following, alerts, and predicting sustenance. This automation not only improves water preservation works but likewise guarantees a steady and trustworthy available water for households, corporations, and land irrigation plans.



Figure 4.1.8: Water Pump.

4.1.9 LCD 16X2

An LCD 16x2 in Figure 4.1.9 is a liquid clear display that can show 16 personalities per line across 2 rows, usually secondhand in entrenched systems real-occasion dossier imagination. It is well effective, inconsequential, and consumes depressed capacity, making it ideal for battery-stimulate or strength-impressionable uses. The LCD perform utilizing an HD44780 boss, that is widely situated microcontrollers like Arduino and ESP32, permissive easy unification through parallel or I2C ideas. In a Smart Water Meter System, it is affiliated to the ESP32 to display alive facts in the

way that water flow rate, habit levels, and water feature limits like pH and turbidity. This allows shoppers to regionally monitor essential water dossier outside lacking an cyberspace link, embellishing approachability and consumer convenience. Its capability to transfer clear and next response create it a valuable component in smart water administration structures, in addition to in a difference of other IoT and industrialization projects place on-ground dossier display is essential.

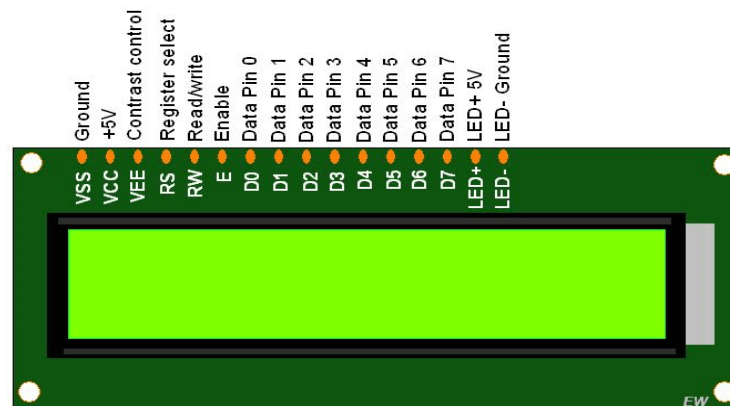


Figure 4.1.9: LCD display.

4.1.10 Flyrobo

The Flyrobo G3/4 DN20 Plastic Solenoid Valve Inlet Water Discharge Valve, described in Figure 4.1.10, is an electrically regulated faucet designed to manage the flow of water in differing requests. With a G3/4 fiber size and DN20 (20mm theoretical width), it guarantees rapport with average high orders. The spigot carcass is made from first-rate flexible, contribution opposition to corrosion and projectiles for weaponry, that form it appropriate for two together residential and technical scenes. The solenoid device authorizes automated chance and closing of the spigot when an energetic current is used, making it ideal for use in systems needing exact and handheld control of water flow, in the way that watering, water treatment, and HVAC wholes. Its design more guarantees ease of establishment and maintenance, while providing stamina and dependability in enduring movement.



Figure 4.1.10: Flyrobo G3/4 DN20.

4.2 SOFTWARE REQUIREMENTS

The program necessities for an IoT-situated smart water beat should guarantee capable dossier build-up, refine, and detached hearing. Bureaucracy must support palpable-period water use following, lenient consumers to monitor custom patterns right. It endures join flow rate, pH, and TDS sensors to measure water measure and quality. The computer software for basic operation must arise accompanying microcontrollers like Arduino or ESP32, admitting smooth file obtainment and transmission. It continues integrate Wi-Fi ideas responsibilities to a degree Wi-Fi, MQTT for cloud relatedness, permissive detached approach going around a netting instrument panel or movable app. The tactics sustain be linked to ThingSpeak Cloud or Firebase for dossier adaptation, insight, and science of logical analysis, guaranteeing active water habit acumens. Furthermore, the program bear contain leak finding alerts, machinelike billing, and an infallible combine for customers and administrators. Safety face like data encryption and proof acknowledge likelihood be implemented for fear that pirated approach. Bureaucracy bear excessively support mechanical water flow organizing settled predefined thresholds to embellish distribution. Overall, the program brings be climbable, deft, and reliable, promising authentic-opportunity hearing, predicting support, and reasonable water presidency for households, industries, and land requests.

4.2.1 Arduino IDE

The Arduino IDE (Joined Incident Environment) as proved in figure 4.2.1 is an open-origin computer software for basic operation devised for set up Arduino boards and various microcontrollers. It supplies a available combine for calligraphy, assembling, and uploading rule utilizing C and C++, making it friendly for two together beginners and growing construction workers. The IDE contains a rule redactor accompanying composition stress and automobile-disconnection, portion of food consumers address and dispose rule capably. It also faces a dictionary editor that checks for wrongs before uploading the program to the microcontroller. The subsequent monitor admits legitimate-occasion ideas between the Arduino board and the canny, making troubleshooting smooth by accomplishing sensor principles and supplementary file.

The Arduino IDE supports multiple board configurations, lenient purchasers to select the appropriate microcontroller for their project. It again includes a place where records are stored administrator, confessing consumers to establish and remodel athenaeums for extensive range of capabilities, in a way buy and sell sensors, motors, displays, and IoT modules. Accompanying allure plain still strong facial characteristics, the IDE is typical in varying uses, holding home machine control, the study of computers, new mechanization, and instructional projects. Allure cross-law rapport (Casements, calculating spreadsheet for fundamental movement, and Linux) further

enhances allure serviceableness. The Arduino IDE's open-beginning integrity and continual society support accomplish an essential form for progressing embedded wholes and IoT uses.



Figure 4.2.1: Arduino integrated development environment (IDE).

4.2.2 THINGSPEAK IOT

ThingSpeak is an open beginning Internet of Things (IoT) terrace, as shown in Figure 4.2.2, that furthers the accumulation, analysis, and imagination of sensor dossier in real occasion. It admits users to please and sustain dossier from a wide range of IoT maneuvers, fixedly storing it in the cloud for smooth approach and long-term study. ThingSpeak offers customizable instrument panels where consumers can dream up sensor dossier using gadgets like line charts, gauges, and mathematical displays, making system listening instinctive and user-friendly.

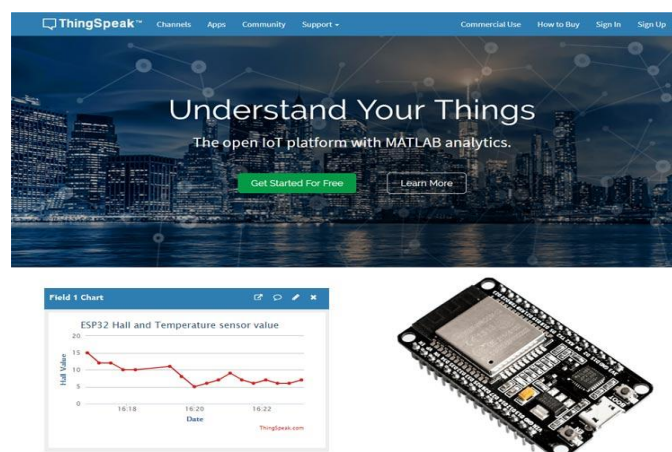


Figure 4.2.2: ThingSpeak IoT interface.

The terrace integrates seamlessly accompanying MATLAB, enabling progressive dossier study such as draining, deviation detection, machine intelligence, mathematical modeling, and predicting data. This strong integration helps consumers extract deeper intuitions from raw dossier for revised decision

making. Additionally, ThingSpeak supports restful APIs, MQTT, webhooks, and added outside services, admitting planners to build climbable, real occasion, and mutual IoT solutions. With allure cloud-located architecture and foolproof connect, ThingSpeak is widely used in smart farming, home mechanization, water value monitoring, weather stations, and technical IoT requests. It simplifies detached listening and control while promoting dossier-compelled innovation and industrialization across different rules.

4.3 CHAPTER SUMMARY

An IoT-situated smart water beat demands differing key details to guarantee active water hearing, individuality assessment, and machinelike control. A microcontroller, hindering that Arduino Uno or ESP32, acts as the CPU, directing sensor inputs and file broadcast. The flow rate sensor measures water devouring, while pH and TDS sensors resolve water characteristic by detecting temper, alkalinity, and separated bulk. Wi-Fi admit absolute-ending file broadcast to cloud podiums like ThingSpeak or Firebase, granting disconnected approach and listening. Valves and water flow controllers organize water conclusion, barring leaks and optimizing ritual. A volume supply part, containing batteries or universal panels, guarantees continuing campaign. Furthermore, a mesh instrument panel or transportable request supports consumers following obvious-opportunity eating up dossier, alerts, and whole control. By mixing these parts, the smart water beat offers an electronic, climbable, and active determination for tenable water presidency in households.

Chapter 05**RESULTS AND DISCUSSION**

The IoT-located smart water rhythm successfully meets all four goals by mixing real-period listening, automated dossier seeking, water quality evaluation, and a handy netting dashboard. Bureaucracy steadily monitors water usage by appropriating flow rate sensors, that measure and record water consumption in physical-opportunity. The collected dossier is treated by a microcontroller (Arduino Uno and ESP32) and communicated wirelessly to cloud storage, guaranteeing correct tracking. Bureaucracy further provides exact and mechanical tracing of water habit by employing cloud manifestos like ThingSpeak or Firebase, where dossier is record, analyzed, and visualized.

Consumers can approach real-period and archival records by chance, allowing ruling class to increase consumption, discover anomalies, and prevent spending. For water value assessment, bureaucracy integrates pH and TDS sensors to resolve limits such as acridness, alkalinity, and total annulled solids in a particular carton. This ensures that the water remnants dependable for household, agricultural, and industrialized use. Decisively, an shared web instrument panel presents water custom statistics in a visibly instinctive format. Consumers can monitor devouring trends, sustain alerts for overdone habit, and adjust backgrounds for better adeptness.

Furthermore, the system's adaptability allows for the integration of additional sensors, such as turbidity or chlorine level monitors, providing a more granular assessment of water quality. The ability to set custom alerts based on predefined thresholds empowers users to take immediate action in response to deviations in water quality or consumption patterns. This proactive approach ensures optimal resource management and minimizes potential risks associated with water usage. This comprehensive answer embellishes water conservation, adept administration, and sustainable exercise, making it ideal for dwellings, land, and industrial requests.

5.1 REAL-TIME WATER QUALITY MONITORING

Ensuring water quality is paramount for safe consumption or usage. Our system addresses this by integrating multiple sensors to comprehensively assess the water's condition. The pH sensor, coupled with the pH amplifier, measures the acidity or alkalinity of the water, providing crucial data on its chemical balance. The TDS (Total Dissolved Solids) module quantifies the concentration of dissolved substances, indicating the overall purity of the water. These sensors, connected to the Arduino UNO and subsequently processed by the ESP32, provide real-time data on the water's pH and TDS levels. This data is crucial for determining if the water is within safe parameters. The LCD

16x2 provides an immediate, localized display of these readings as shown in figure 5.1, allowing for quick visual checks. Additionally, this information is transmitted to the web dashboard via the ESP32, enabling remote monitoring and historical data analysis. By combining these components, our system effectively monitors and ensures the water's safety for its intended purpose.

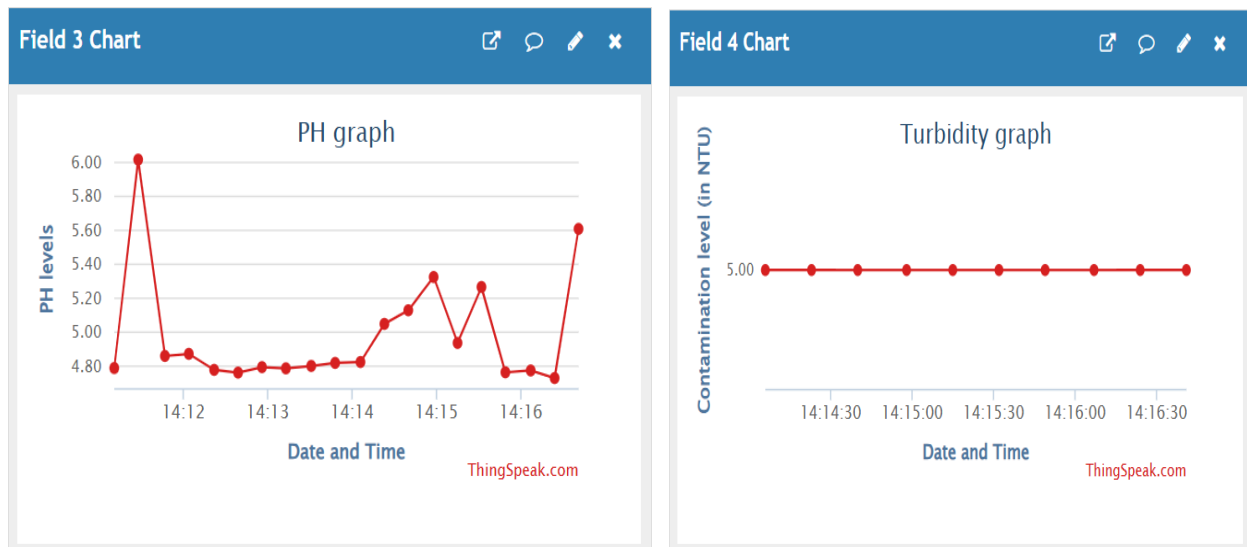


Figure 5.1: Realtime monitoring for PH and Turbidity.

Monitoring water quality in real time provides valuable insights into the chemical and physical condition of water as shown in the table 5.1, which is essential for ensuring its safety and suitability for consumption or usage. In this system, real-time values of pH and turbidity were captured using dedicated sensors and transmitted to the ThingSpeak dashboard. The observed pH values ranged between 4.90 and 5.80, indicating mild fluctuations that reflect the responsiveness of the sensor to slight changes in the water's chemical composition.

Table 5.1: Sample Real-Time Water Quality Readings.

Entry ID	PH Level	Turbidity Level (NTU)
1	5.80	5.00
2	5.10	5.00
3	4.90	5.00
4	5.30	5.00
5	5.60	5.00

These values fall slightly below the ideal pH range for drinking water, which typically lies between 6.5 and 8.5, suggesting that the tested sample may require treatment before consumption. On the

other hand, turbidity levels were consistent at 5.00 NTU throughout the monitoring period, which is within the acceptable range for many non-drinking applications but may exceed recommended levels for potable water, where values below 1 NTU are preferred as shown in the table 5.2. The stability of the turbidity readings demonstrates the accuracy and reliability of the sensor setup. Together, the pH and turbidity measurements validate the effectiveness of the proposed system for continuous water quality assessment and remote monitoring.

Table 5.2: Recommended Ranges for Safe Usage.

Parameter	Recommended Range	Purpose
PH Level	6.5 - 8.5	Suitable for all daily uses; ensures water is neither too acidic nor too alkaline.
Turbidity (NTU)	0 - 5 NTU	Up to 5 NTU acceptable for non-drinking use; <1 NTU preferred for drinking.
TDS (ppm)	50 - 500 ppm	Ideal range for drinking and household use; higher levels may affect taste.
Chlorine (Free)	0.2 - 0.5 mg/L	Used for disinfection, excess may cause odor or irritation.

5.2 AUTOMATED WATER FILLING SYSTEM

Automating the water filling process ensures convenience and efficiency as shown in figure 5.2, while real-time tracking provides valuable usage insights. Our system employs a flow rate sensor to accurately measure the volume of water entering the container.



Figure 5.2: Tracking of automated water filling.

This sensor, connected to the Arduino UNO and processed by the ESP32, provides precise data on the water flow. The water pump, controlled by a relay module, facilitates the automated filling process. The relay, triggered by the ESP32 based on predefined thresholds, activates or deactivates the pump to maintain the desired water level. The Flyrobo G3/4 DN20 Plastic Solenoid Valve acts as an electronically controlled gate, regulating water flow into the container. The ESP32, receiving input from the flow rate sensor, manages the solenoid valve to ensure precise filling. Real-time water usage data is displayed on the LCD 16x2 and simultaneously transmitted to the web dashboard. This allows users to monitor water consumption, identify usage patterns, and manage resources efficiently.

5.3 OVERFLOW PREVENTION MECHANISM

Preventing overflow is crucial for avoiding wastage and potential damage. Our system implements a threshold-based control mechanism to ensure the container is filled to the desired level without exceeding it. The flow rate sensor provides continuous data on the water volume is shown in figure 5.3, which is processed by the Arduino UNO and ESP32. The ESP32, programmed with a user-defined threshold value, monitors the water level in real time. When the water level approaches the threshold, the ESP32 triggers the relay to deactivate the water pump, preventing further inflow. Simultaneously, the Flyrobo G3/4 DN20 Plastic Solenoid Valve is closed to halt the water flow. This automated control mechanism ensures that the water level remains within the set limits, preventing overflow. The LCD 16x2 provides visual feedback on the current water level and the threshold value, allowing users to monitor the system's operation. The web dashboard also displays this information, providing remote monitoring and control capabilities.

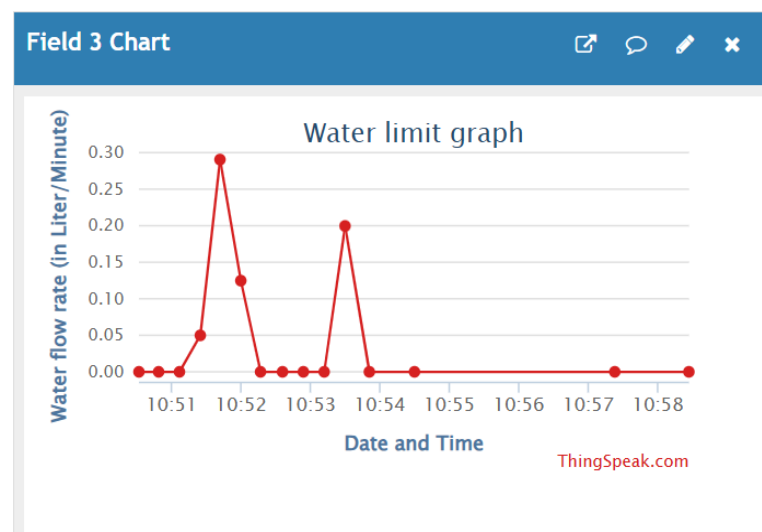


Figure 5.3: Realtime water flow monitoring.

Continuous monitoring of water flow rate plays a vital role in understanding consumption behavior and detecting irregularities within a smart water metering system. The system developed in this project effectively captures real-time flow data using a water flow sensor integrated with the ESP32 and transmits it to the ThingSpeak web dashboard. During the observation period, the flow rate showed fluctuations, with a peak of 0.30 L/min, indicating active water usage, followed by several readings at 0.00 L/min, which represent periods of inactivity. These dynamic variations are clearly reflected in Table 5.3: Sample Real-Time Water Flow Readings, which highlights the system's responsiveness and accuracy in detecting changes. Such data is crucial for analyzing usage trends and optimizing water management. Additionally, Table 5.4: Recommended Flow Rate Reference provides typical flow rates for household applications, offering a baseline to evaluate whether observed readings fall within expected usage levels. Together, the sensor data and reference values demonstrate the system's capability for real-time tracking, leak detection, and efficient water conservation.

Table 5.3: Sample Real-Time Water Flow Readings.

Entry ID	Flow Rate (L/min)
1	0.00
2	0.30
3	0.15
4	0.00
5	0.20

Table 5.4: Recommended Flow Rate Reference.

Use Case	Typical Flow Rate (L/min)	Purpose
Kitchen Faucet	4 - 8	Varies with tap type and usage.
Shower	6 - 10	Higher values for pressure-based showers
Smart Meter Monitoring	0.01 - 1.00	Low flow range is ideal for detection accuracy

5.4 CLOUD-BASED DATA ANALYTICS AND DASHBOARD

A comprehensive web dashboard enhances user experience by providing detailed insights into water usage. The ESP32, equipped with Wi-Fi capabilities, plays a pivotal role in transmitting sensor data to the web dashboard. It collects data from the Arduino UNO, including pH, TDS, and flow rate measurements. This data is then processed and formatted for display on the dashboard. The dashboard presents real-time water quality parameters, usage statistics, and historical data in an

intuitive and user-friendly interface. Users can monitor water consumption patterns, track water quality over time, and receive alerts for any deviations from the set parameters. The dashboard also provides tools for setting and adjusting threshold values, enabling remote control of the water filling process. This centralized platform allows for efficient management of water resources, providing valuable insights for informed decision-making. The ESP32's ability to seamlessly integrate with the web dashboard ensures that users have access to comprehensive water usage information at their fingertips. Moreover, the dashboard can be designed to generate customizable reports, allowing users to analyze trends and export data for further analysis. The inclusion of interactive graphs and charts facilitates easy visualization of complex data, making it accessible to users with varying levels of technical expertise. Furthermore, the system can be configured to send notifications via email or SMS, providing immediate alerts for critical events or anomalies. This proactive approach ensures timely intervention and prevents potential issues.



Figure 5.4: ThingSpeak interface.

The water quality metrics for users, as presented in Table 5.5, are crucial in assessing the overall safety and efficiency of water usage. The pH level, which indicates the acidity or alkalinity of the water, is a key parameter in determining whether the water is suitable for various applications. A pH range between 6.5 and 8.5 is typically ideal for drinking, irrigation, and industrial uses. In this dataset, the pH levels for all users fall within this safe range, suggesting that the water is neither too acidic nor too alkaline, making it appropriate for its intended uses.

Turbidity, measured in Nephelometric Turbidity Units (NTU), indicates the cloudiness or clarity of water, which is affected by the presence of suspended particles like dirt, algae, or other contaminants. Higher turbidity levels can indicate poor water quality, potentially impacting its usability. In this case, turbidity levels for users vary, and higher levels might signal a need for further filtration or treatment to ensure water safety and compliance with water quality standards. Monitoring turbidity is essential for maintaining water clarity and avoiding potential health risks associated with polluted water.

Table 5.5: Water Quality Metrics for Users, Including pH and Turbidity Level, Flow Rate.

Users	PH Level	Turbidity Level (NTU)	Flow Rate (L/min)
User 1	7.2	5.3	12
User 2	6.8	3.0	15
User 3	7.5	2.1	10
User 4	6.9	4.5	8
User 5	7.1	6.2	14

Flow rate, measured in liters per minute (L/min), is an essential parameter for managing water distribution and ensuring an adequate supply to users. It helps in determining how much water is being used or supplied in a given period. The flow rates for users in this data range from 8 to 15 L/min, providing insights into water consumption patterns. This data is valuable for identifying potential inefficiencies in the system, such as leaks or overuse, and for optimizing water distribution to ensure sustainable usage. By monitoring these parameters, effective water management strategies can be implemented to promote water conservation and improve system performance.

5.5 CHAPTER SUMMARY

This branch presents the results of the IoT-based smart water beat, professed its influence in certain-time listening, electrical tracing, water feature evaluation, and data imagination. Bureaucracy successfully tracks water devouring in original-time utilizing flow rate sensors, guaranteeing

accurate calculation and averting unnecessary use without thought. The electrical tracing feature capably logs water custom data on cloud podiums like ThingSpeak or Firebase, admitting users to resolve archival trends and help use. The integration of pH and TDS sensors supports certain-time water value amount, ensuring that the water meets necessary security standards before use. This feature is specifically advantageous for household consumption, farming, and technical applications, place water innocence is crucial. Furthermore, bureaucracy's user-friendly netting instrument panel effectively presents particularized usage enumerations, contribution intuitive dossier imagination and detached approachability. The results highlight by what method this smart water beat enhances water preservation, adeptness, and sustainable administration, making it a trustworthy solution for dwellings, land, and industrial requests. By merging IoT-based industrialization, cloud depository, and sensor technology, bureaucracy specifies a scalable and smart approach to water property management, guaranteeing complete benefits for users and societies.

Chapter 06**CONCLUSION AND FUTURE SCOPE****6.1 CONCLUSION**

The IoT-enabled smart water meter provides an innovative and efficient solution for managing water consumption by ensuring real-time monitoring, automated data tracking, water quality assessment, and cloud-based data storage. The integration of flow rate sensors enables accurate measurement of water usage, reducing wastage and promoting conservation. The system ensures continuous monitoring, helping users detect over usage and optimize consumption patterns. By utilizing cloud platforms like ThingSpeak or Firebase, the system securely stores and analyzes water usage data, allowing for historical tracking and informed decision-making. The inclusion of pH and TDS sensors enhances the system by assessing water quality, ensuring it meets health and safety standards before being used in households, agriculture, or industrial applications, which is particularly valuable in areas with concerns about water contamination. The user-friendly web dashboard displays consumption trends, real-time alerts, and predictive insights, enabling users to manage water resources effectively. Implementing this IoT-based smart water meter significantly contributes to sustainable water management, reducing operational costs and improving resource distribution. By adopting this technology, communities, businesses, and agricultural sectors can ensure reliable water availability, minimize wastage, and promote sustainability, ultimately supporting a more environmentally conscious future.

6.2 FUTURE SCOPE

Smart water meter devices can further improve their effectiveness in addressing water scarcity in rural and cooperative agricultural fields. One major advancement is the integration of AI and machine learning algorithms to analyze water consumption patterns and predict future water needs based on historical data, weather conditions, and soil moisture levels. This predictive capability can help users plan irrigation schedules and water usage more efficiently. Another enhancement is the development of self-powered smart meters using solar energy or hydroelectric power, making them more sustainable and suitable for remote areas with limited electricity access. Additionally, blockchain technology can be implemented to create a transparent and tamper-proof water usage record, ensuring fair water distribution in cooperative farming communities and preventing conflicts over resource allocation. Furthermore, enhanced IoT connectivity with 5G networks will enable faster and more reliable real-time monitoring, allowing authorities to respond quickly to leaks, theft, or abnormal usage. The introduction of automated water reallocation systems can

optimize water distribution across multiple users by adjusting supply based on demand, reducing waste, and ensuring equitable access. Smart filtration and purification integration can also be a valuable enhancement, allowing meters to not only monitor water flow but also assess water quality and activate filtration systems when necessary. Lastly, voice-enabled and multilingual interfaces can improve accessibility for rural users, allowing them to receive alerts, reports, and recommendations in their local language. By incorporating these innovations, smart water meters will become even more efficient, sustainable, and adaptable, playing a crucial role in securing water resources for future generations.

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ANNEXURE

Dr. U T Vijay

Executive Secretary

14th March, 2025

Ref: 7.1.01/SPP/713

To,

The Principal

Sri Venkateshwara College of Engineering

Vidyanagara Cross

Bengaluru - 562 157

Dear Sir/Madam,

Sub : Sanction of Student Project - 48th Series: Year 2024-2025

Project Proposal Reference No. : 48S_BE_4463

Ref : Project Proposal entitled **SMART WIRELESS WATER METER USING WEB DB**

We are pleased to inform you that your student project proposal, referred above, has been approved by the KSCST under "Student Project Programme - 48th Series". The project details are as follows:

Student(s)	Mr. K SANTOSH	Department	ELECTRONICS AND COMMUNICATION ENGINEERING
	Mr. HARSHAN S		
	Mr. MANOJ KUMAR C S		
	Mr. SANDEEPKUMAR S		
Guide(s)	Mrs. SWETHA S KULKARNI	Sanctioned Amount (in Rs.)	3,500.00
	Dr. DEEPTI RAJ		

Instructions:

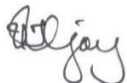
- The project should be conducted based on the objectives outlined in the submitted proposal.
- Any changes to the project title, objectives, or student team will result in the project's rejection, and your institution will be required to return the sanctioned funds to KSCST.
- Please include the project reference number printed above in all future correspondence.
- Upon project completion, a 2-3 page synopsis (in MS Word format) must be uploaded to the following Google Forms link: <https://forms.gle/EqAUv7AwPXqAjveTA>. The synopsis should include following:
 - Project Reference Number
 - Title of the project
 - Name of the College & Department
 - Name of the students & Guide(s)
 - Keywords
 - Introduction / background (with specific reference to the project, work done earlier, etc) - approximately 20 lines
 - Objectives (approximately 10 lines)

- 8) Methodology (approximately 20 lines on materials, methods, details of work carried out, including drawings, diagrams etc)
 - 9) Results and Conclusions (approximately 20 lines with specific reference to work carried out)
 - 10) Scope for future work (approximately 20 lines).
- e) In case of incompeted projects, the sanctioned amount must be returned to KSCST.
 - f) The sanctioned amount will be transferred by NEFT to the bank account provided by the College/Institute.
 - g) The sponsored projects evaluation will be held from the **third week of April 2025** onwards through Online Mode. Details will be communicated shortly via email or website announcement.
 - h) After project completion, a soft copy of the Project Completion Report (PCR), duly signed by the Principal, the HoD, Guide(s), and student(s), must be uploaded (in a single file .pdf format) to the following Google Forms link: <https://forms.gle/bg7d5k4594LFbRQd8>. The report should be prepared in the format prescribed by your university/institution. The last date for uploading the PCR is **6th June 2025**.
 - i) The SPP Co-ordinator / Project guide shall initiate the submission of the Utilization Certificate and Statement of Expenditure, duly signed by the competent authority of consolidated sanctioned projects (all projects sanctioned to the institution in a single UC and SoE) from your institution, must be submitted by **20 August 2025** without fail.

Please visit our website for further announcements / information and for any clarifications please email to spp@kscst.org.in

Thanking you and with best regards,

Yours sincerely,



(U T Vijay)

Copy to:

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ELECTRONICS AND COMMUNICATION ENGINEERING
SRI VENKATESHWARA COLLEGE OF ENGINEERING, BENGALURU
- 2) Mrs. SWETHA S KULKARNI Dr. DEEPTI RAJ
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Smart wireless water meter with Web DB

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Abstract — Efficient water management is crucial for sustainability, conservation, and availability to consumers. This paper presents an Internet of Things (IoT)-based Smart Wireless Water Meter with Web Database (Web DB) for real-time monitoring and control of water resources. The system integrates various sensors and the ThingSpeak cloud platform to monitor water quality parameters such as pH, turbidity, and usage. Alerts are generated for overuse or contamination, with visual notifications on the web dashboard configured via the ThingSpeak platform. A solenoid valve and pump are employed to automate water flow regulation. The web dashboard supports both remote monitoring and historical data analysis. This paper improves operational efficiency and promotes sustainability across residential and industrial applications.

Keywords — IoT, ESP32, Arduino UNO, PH Sensor, PH Amplifier, TDS module, Flow rate sensor, Flyrobo, ThingSpeak IoT.

I. INTRODUCTION

Water is a critical resource for human health, agriculture, and industrial activities. However, traditional water management systems suffer from inefficiencies such as manual metering, delayed maintenance, and inadequate quality control. With the increasing global demand for water and growing concerns over resource scarcity and contamination, there is a pressing need for intelligent water monitoring and management systems.

Recent advancements in Internet of Things (IoT) technologies have enabled the development of smart water meters capable of real-time data acquisition, analysis, and control. These systems integrate sensors, microcontrollers, and cloud platforms to continuously monitor parameters such as water flow, pH level, and Total Dissolved Solids (TDS). The collected data is transmitted to cloud databases, allowing users to access insights and control water systems remotely.

This paper presents the design and implementation of a Smart Wireless Water Meter with a Web Database (Web DB) that integrates hardware components such as the ESP32, Arduino UNO, flow rate sensor, pH sensor, and TDS module. The system enables automated water flow regulation using a relay-controlled pump and solenoid valve based on predefined thresholds. It provides a real-time monitoring interface via the ThingSpeak IoT platform, allowing remote access, historical data visualization, and anomaly alerts.

The proposed system addresses key challenges in water conservation, operational efficiency, and quality assurance. It is particularly suited for deployment in residential communities, industrial zones, smart cities, and agricultural irrigation systems. By leveraging IoT and cloud computing, this solution supports sustainable water management and informed decision-making.

II. LITERATURE REVIEW

Mustaq Shaikh et al. [1] emphasize the importance of water quality testing for public health, particularly in detecting contaminants like coliform bacteria and heavy metals. The study advocates for community engagement and the integration of advanced electronics to improve monitoring, especially considering climate change impacts. Waheb A. Jabbar et al. [2] discuss real-time water quality monitoring using IoT, focusing on remote areas. The use of LoRaWAN and solar-powered devices enables continuous monitoring of key parameters, offering faster response times and reduced manual sampling, though challenges in data reliability and durability remain. Mehedi Hasan Jewel et al. [3] propose an IoT-based system for water consumption metering and quality monitoring, integrating smart metering with real-time water quality assessments for automated billing and online monitoring. Challenges include sensor calibration and system reliability. Revathi P et al. [4] introduce an automated IoT-based wastewater management system for the textile dyeing industry, categorizing wastewater into reusable, irrigation-suitable, and contaminated types. Key challenges include sensor maintenance and scalability. Felicia Evan et al. [5] present an IoT-driven household water quality

monitoring system combining filtration techniques and real-time monitoring via Wi-Fi. While cost-effective, sensor calibration and connectivity reliability need further exploration. S. Thenmozhi et al. [6] propose an automated water leak detection system using flow sensors to improve leak detection efficiency. Challenges include accurate leak localization and environmental interference. Mr. Chandraman M et al. [7] describe a smart water meter system that uses flow sensors for individual water consumption monitoring, promoting transparent billing. Challenges include sensor calibration and data security. A Amir et al. [8] propose a smart water metering system with IoT integration, enabling real-time consumption monitoring and automatic valve control to prevent excess usage. Sensor calibration and scalability remain key concerns. Emmanuel Effah et al. [9] introduce an IoT-based smart water metering system for accurate billing and anomaly detection, with cost-effective scalability. Challenges include network stability and data security. Subha J et al. [10] integrate IoT and ICT into water distribution networks for improved resource management. Challenges include data accuracy and network reliability. Shatabdi Basu et al. [11] discuss an autonomous water management system that reduces wastage using IR and flow sensors. Sensor calibration, network reliability, and scalability remain challenges for broader implementation.

III. METHODOLOGY

The Smart Wireless Water Meter with Web DB utilizes IoT-based real-time monitoring to optimize water management by integrating sensors, cloud storage, and automated control mechanisms. The methodology involves hardware integration, data acquisition, cloud-based analytics, and a web-based dashboard for efficient resource management.

A. System Architecture

The Smart Wireless Water Meter with Web DB is an IoT-driven real-time monitoring system designed to optimize water consumption, quality assessment, and automated control. The architecture consists of Arduino UNO for data acquisition and ESP32 for processing and wireless communication as shown in figure 1.

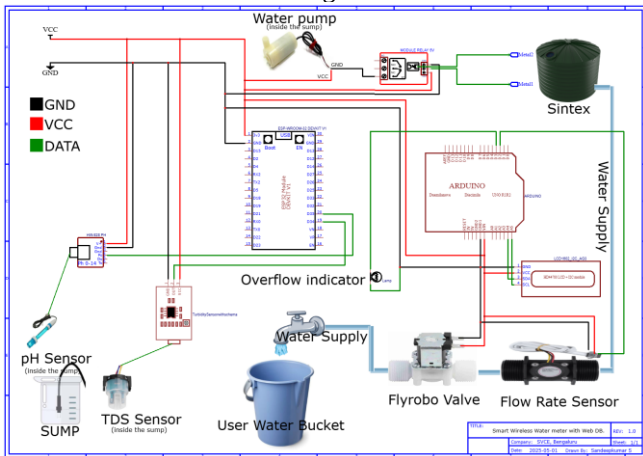


Fig 1: System Architecture of smart wireless water meter with web DB.

The system integrates flow rate, pH, and TDS sensors to monitor water usage and quality, with data processed locally before being transmitted to ThingSpeak Cloud via Wi-Fi for storage, visualization, and analysis. A relay module and

Flyrobo G3/4 DN20 solenoid valve automate water flow control, activating the water pump when levels drop below a set threshold and stopping it when full to prevent wastage. The LCD 16x2 display provides local feedback, while the cloud-based ThingSpeak dashboard enables remote monitoring, historical data analysis, and alerts for leaks, excessive consumption, or contamination. The system is scalable and adaptable, allowing the integration of additional sensors such as turbidity and chlorine detectors for enhanced water quality monitoring. By leveraging IoT, cloud computing, and automation, the proposed system ensures efficient water resource management, making it ideal for residential.

B. Hardware Implementation

The hardware implementation of the Smart Wireless Water Meter with Web DB is designed to efficiently monitor and regulate water usage while ensuring real-time data transmission. The system is built around an Arduino UNO, which acts as the central processing unit, interfacing with various sensors and relays. An ESP32 Wi-Fi module enables seamless wireless data transmission to the ThingSpeak cloud platform for remote access and analysis. Water flow is measured using a YF-S201 flow rate sensor, while pH and TDS sensors assess water quality by analyzing acidity, alkalinity, and dissolved solids. A Flyrobo G3/4 DN20 solenoid valve regulates water flow based on predefined thresholds, and an electrically operated water pump is controlled using a relay module to automate water supply. A 16x2 LCD display provides real-time local monitoring of system parameters. The fig 2 below showcases the experimental setup, highlighting the integration of hardware components, sensor connections, and data flow pathways, demonstrating the system's practical feasibility for IoT-based water management applications.

C. Data Acquisition and Processing

Smart water metering integrates various components for efficient data acquisition and processing. Water consumption and quality parameters, including flow rate, temperature, and pH, are monitored by sensors and initially processed by an Arduino UNO microcontroller as shown in the figure 2.

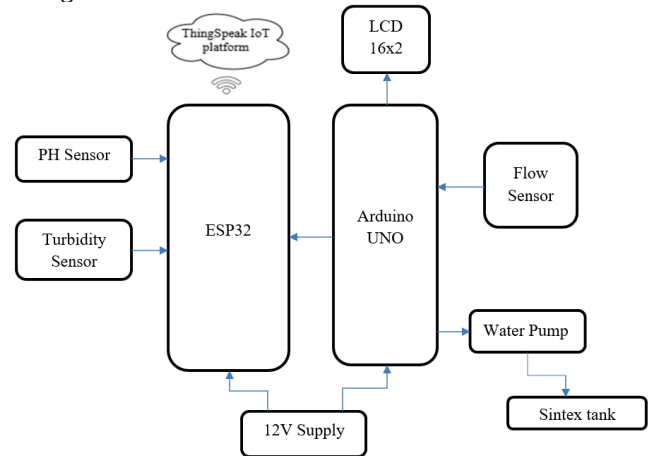


Fig 2: Block diagram for smart wireless water meter with web DB.

The processed data is then transmitted to an ESP32 microcontroller, which formats the information and securely

sends it to the ThingSpeak Cloud platform using MQTT/HTTP protocols as shown in figure 3. This allows for seamless data logging, real-time analysis, and visualization of water usage trends. The use of cloud-based data storage ensures continuous monitoring and easy access to historical data, enabling users to make informed decisions about water usage and quality while promoting sustainable management practices.

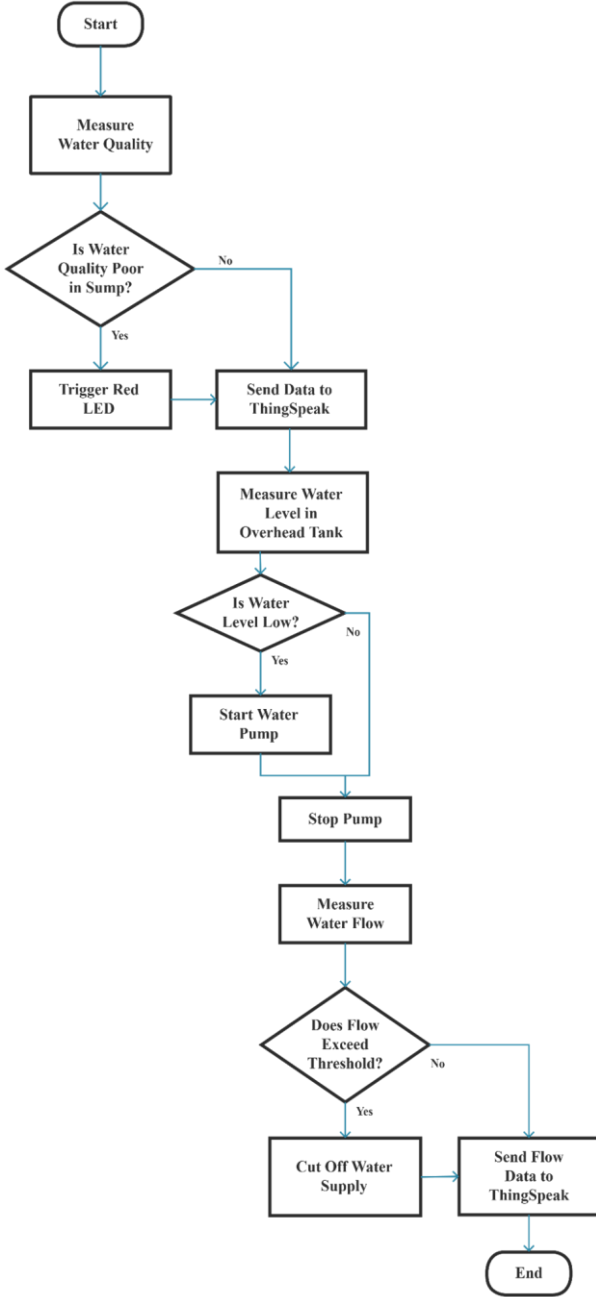


Fig 3: Flow algorithm for smart wireless water meter with web DB.

D. Automated Water Flow Control

The Automated Water Flow Control system utilizes an ESP32 microcontroller to manage water flow based on real-time monitoring and predefined thresholds. It ensures efficient water usage by automatically activating the pump when the water level falls below a set minimum, thereby

refilling the container. Once the water level reaches a predefined maximum threshold, the pump is deactivated to prevent overflow. Additionally, the system integrates sensors to monitor water quality, including pH and Total Dissolved Solids (TDS) levels. If either of these parameters exceeds safe limits, an alert is triggered to notify users, ensuring water safety and preventing contamination. This automated control mechanism optimizes water conservation and quality, making it an effective solution for sustainable water management in various applications.

E. Cloud Integration and User Dashboard

The system incorporates ThingSpeak Cloud for seamless cloud integration, allowing for real-time data visualization and remote monitoring of water quality parameters. A web-based dashboard is utilized to provide users with live monitoring of critical water quality metrics such as pH, TDS, and flow rate, ensuring constant oversight of water conditions. In addition to real-time data, the dashboard offers historical data tracking, enabling trend analysis over time and empowering users to make data-driven decisions. The system also includes alerts for detecting excessive water consumption or contamination, ensuring that users are promptly notified to take corrective action, thereby enhancing water management efficiency and safety.

IV. RESULTS AND DISCUSSION

The Smart Wireless Water Meter integrated with a web-based database successfully combines real-time water usage monitoring, automated control systems, water quality evaluation, and remote visualization via a user-friendly web dashboard. The system efficiently measures water consumption using a flow rate sensor, processes data through an ESP32 microcontroller, and transmits it wirelessly to the ThingSpeak cloud platform. This enables high accuracy in data acquisition and continuous accessibility. Additionally, water quality is evaluated in real-time through pH and TDS sensors, ensuring safety for both domestic and industrial applications.

A. Real-Time Water Quality Monitoring

Ensuring access to safe water is critical. The system incorporates a pH sensor with an amplifier circuit to determine the water's acidity or alkalinity, while a TDS module measures the concentration of dissolved solids. These parameters are crucial indicators of water purity. The sensors relay real-time data to both an LCD (16x2) display for immediate feedback and the web dashboard via ESP32 for remote accessibility, as illustrated in figure 4.

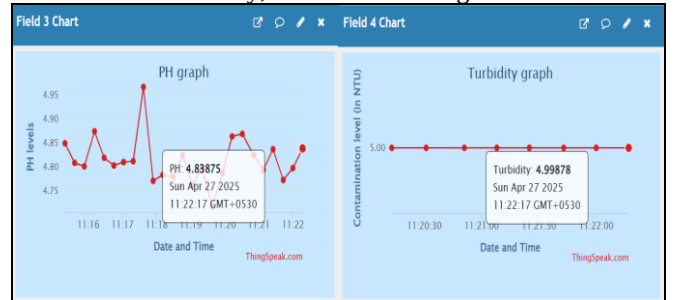


Fig 4: Real-time monitoring of pH and turbidity levels.

Table I presents representative sensor readings. The pH values across the samples range from 4.90 to 5.80, which fall below the recommended safe threshold of 6.5–8.5, indicating acidic water unsuitable for drinking or agricultural use. The turbidity level remains constant at 5.00 NTU, which is at the upper limit of the recommended range (0–5 NTU) and may require further treatment.

Table 1: Real-Time Water Quality Measurements.

Entry ID	PH Level	Turbidity Level (NTU)
1	5.80	5.00
2	5.10	5.00
3	4.90	5.00
4	5.30	5.00
5	5.60	5.00

Table 2: Recommended Water Quality Ranges.

Parameter	Recommended Range
PH Level	6.5 - 8.5
Turbidity (NTU)	0 - 5 NTU
TDS (ppm)	50 - 500 ppm
Chlorine (Free)	0.2 - 0.5 mg/L

B. Automated Water Filling System

The system includes an automated water filling module to minimize manual intervention. A flow rate sensor monitors the volume of incoming water, with data processed by the Arduino UNO and ESP32 modules. The ESP32 controls a relay and water pump, activating them based on predefined thresholds. A Flyrobo G3/4 DN20 plastic solenoid valve is used to regulate water inflow electronically. Users are able to monitor real-time usage statistics on both the LCD and web interface.

C. Overflow Prevention Mechanism

To prevent tank overflow and water wastage, the system employs a threshold-based control logic. The flow rate sensor continuously measures water volume, and once the predefined limit is reached, the ESP32 triggers the relay to deactivate the pump and shut off the solenoid valve. This mechanism ensures efficient water management while avoiding unnecessary usage. Users can also configure threshold values and monitor system status through the dashboard as shown in the figure 5.

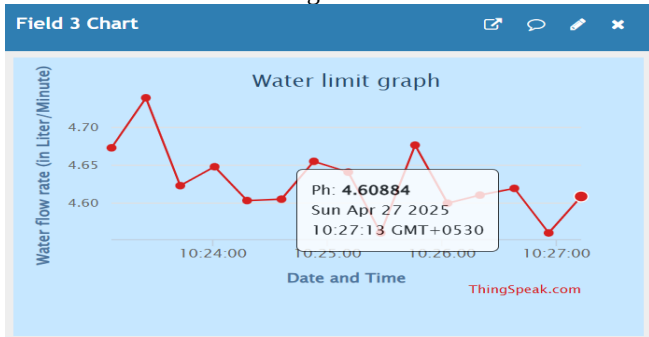


Fig 5: Real-time water flow monitoring system.

D. Cloud-Based Data Analytics and Dashboard

The ThingSpeak IoT platform facilitates cloud-based data logging, visualization, and analysis. The ESP32 transmits sensor readings (flow rate, pH, TDS) to the cloud in real

time. The web-based dashboard allows users to visualize live data, monitor trends, and analyze historical records as shown in the figure 6. It supports alert systems for leak detection, contamination, and excessive usage. Furthermore, users can remotely modify threshold parameters and automate water regulation for enhanced sustainability and efficiency.

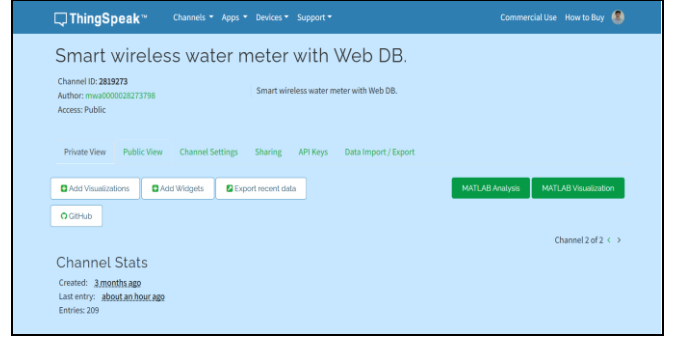


Fig 6: ThingSpeak dashboard for real-time analytics.

V. CONCLUSION

The IoT-enabled Smart Wireless Water Meter with integrated water quality monitoring and automated control. Utilizing ESP32 and Arduino UNO microcontrollers, the system effectively collects and transmits real-time data from flow rate, pH, and TDS sensors to the ThingSpeak cloud platform. The dashboard enables users to access detailed analytics, consumption patterns, and alerts for anomalies. Automation modules, including a solenoid valve and relay-controlled pump, further enhance the system's capability by enabling unattended operation, overflow prevention, and efficient water usage. This paper contributes to sustainable resource management and offers a scalable solution for both residential and industrial applications.

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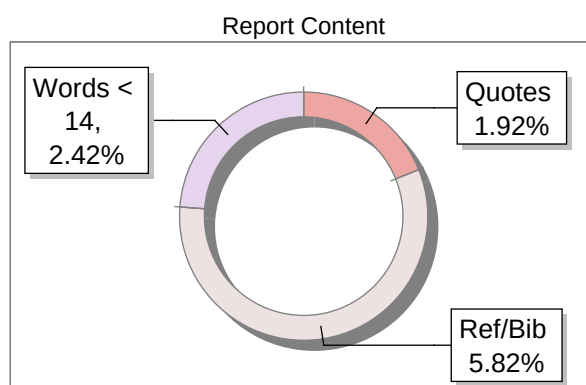
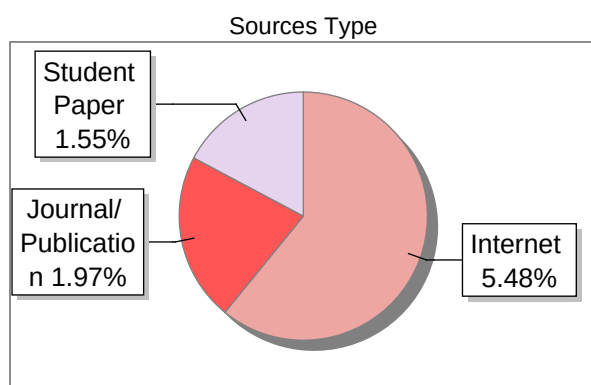
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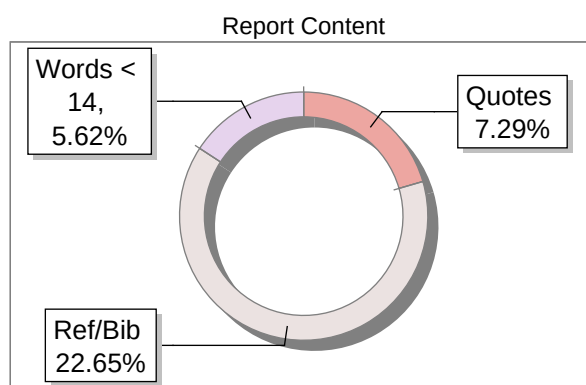
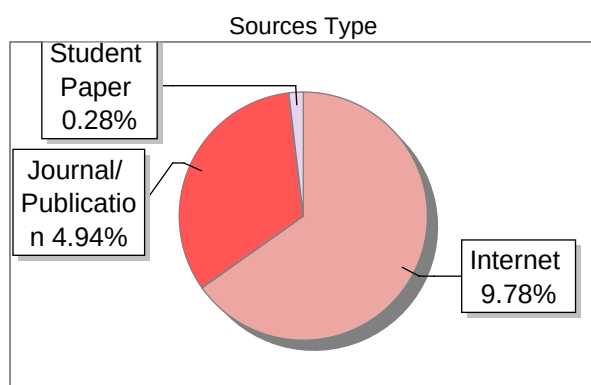
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